Chapter 4
Introduction to Power Device Technology

In the following some basic aspects of power device production technology will be described. A considerable part is devoted to the different methods of doping, which form the heart of semiconductor technology.

4.1 Crystal Growth

The material for silicon devices must be of very high purity. Metallurgical silicon is converted to trichlorosilane $SiHCl_3$ which is liquid, and purified by fractional distillation. Especially chlorides of metals must be eliminated. By reduction of $SiHCl_3$ in a hydrogen atmosphere, polycrystalline rods of pure silicon are formed. For more details, see [Ben99].

The semiconductor material used for power device manufacturing must be monocrystalline. To produce such monocrystals there are two important processes.

In the *Czochralski process* (*CZ*) the crystal growth is performed in a crucible in which molten silicon is kept at a defined temperature. The required p- or n-type dopants are added to the melt. A small monocrystalline seed crystal is then dipped into the melt (Fig. 4.1). During slow rotation of the seed crystal, and of the crucible in the opposite direction, monocrystalline layers of silicon are deposited on the seed crystal maintaining the crystal structure of the seed. The growing bar is simultaneously pulled slowly upwards.

With the CZ process very large single crystals can be grown. Si cylinders of a length of several meters and a diameter of more than 30 cm for fabrication of 300 mm wafers are produced on an industrial scale, and wafers with diameter of 45 cm are introduced [Cla11, Lap08]. However, the purity and quality of single crystals is limited with the CZ-process, since the melt is in contact with the crucible during crystal growth. The oxygen content in CZ-silicon is typically $>10^{17}$ cm^{-3}, and also the content of impurity carbon is in the same range. CZ-wafers are mainly used as substrates for epitaxial wafer growth (see Sect. 4.3), from which integrated

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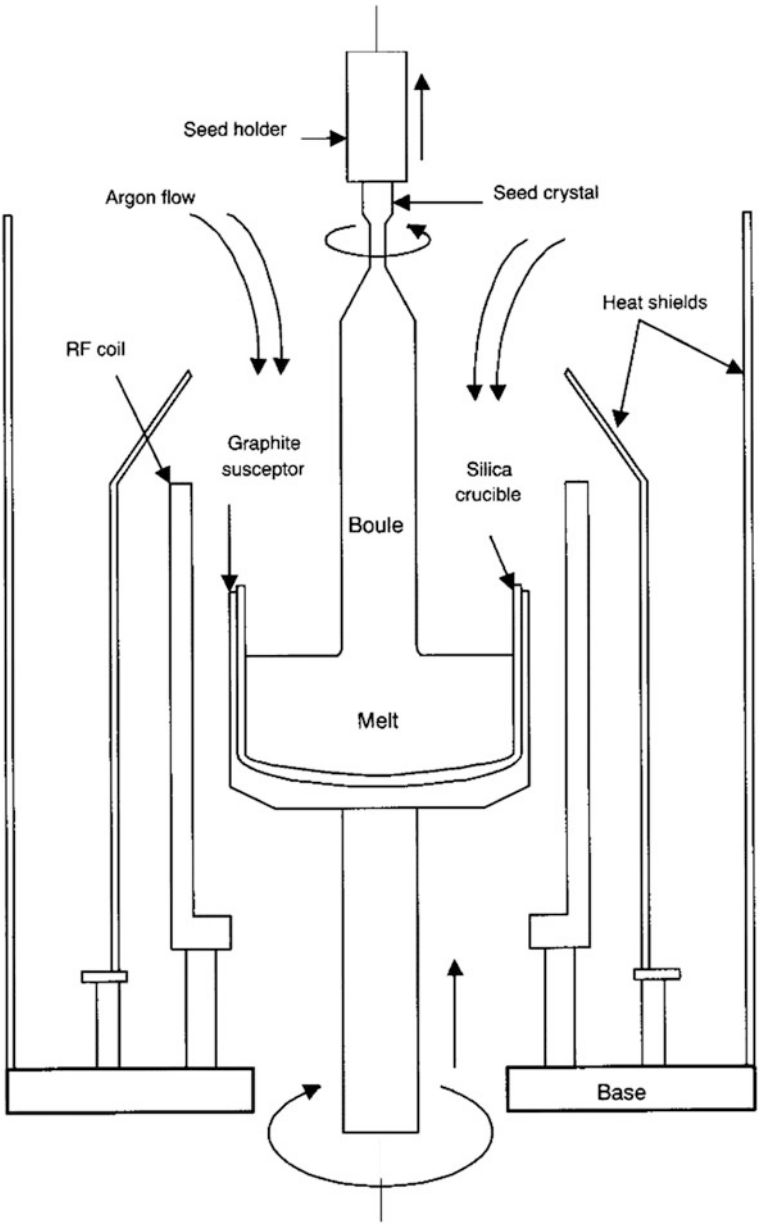


Fig. 4.1 Czochralski-process for production of single crystalline silicon rods. Figure taken from [Ben99]. Reprint with permission of John Wiley & Sons, Inc

circuits etc. are produced. Some power devices like MOSFETs are also produced from epitaxial wafers using CZ-substrates. For power devices in which the volume of the wafer is used, in most cases the purity of CZ-crystals was not sufficient. However, there is strong progress.

Meanwhile, 300 mm magnetic Czochralski Wafers are used for the fabrication of IGBTs [Scu16]. Regarding production effort, this is a large progress. Steady magnetic fields and alternating magnetic fields are used to damp convective flow and to stabilize the conditions of crystal growth [Gal02].

The *Float-Zone (FZ) process* is a crucible free crystal growth method. The starting material is a rod of high purity poly-silicon. A seed crystal is clamped in contact with the end of the rod (Fig. 4.2). An induction heating coil is placed around the rod, melting a small zone next to the seed crystal. While slowly moving the coil along the bar, the molten zone, starting at the interface between seed crystal and poly-silicon rod, passes slowly along the length of the bar. The poly-silicon melts and monocrystalline silicon is regrown with the same orientation as the seed crystal. Because of the absence of a crucible, crystals with higher purity and higher quality can be grown with the FZ process. Also since the solubility of the impurities is higher in the melted zone they tend to be accumulated there. The carbon content is $< 5 \times 10^{15}$ cm^{-3} and the oxygen content is $< 1 \times 10^{16}$ cm^{-3}. With the FZ method, monocrystalline silicon rods with a diameter of up to 20 cm can be produced.

Power devices, which use the whole volume of the wafer, have been mostly fabricated from FZ silicon over a long time because of the higher crystal quality and excellent purity. Most power devices which are manufactured with epitaxial techniques are using CZ silicon as material for the substrates.

In the FZ as well as in the CZ process, the doping of the crystal is done by adding dopants to the melt. Power devices normally require a thick low-doped middle region to support a large voltage. Most often an n-type region is preferred. The value of this doping depends on the voltage for which the power device is designed; see further Eq. (3.84) and Fig. 3.17. The doping processes will be discussed in more detail below.

After crystal growth follows a sawing process to slice the rod in single wafers. To remove the surface damage caused by sawing and to obtain a clean and undisturbed surface, the wafer surface layer is lapped mechanically and in some cases additionally etched chemically. For some semiconductor processes also wafers with single side polished surface are necessary.

4.2 Neutron Transmutation for Adjustment of the Wafer Doping

The homogeneity of the doping concentration is extremely important for power devices. If variations in doping concentration (or local defects) exist in the wafers, the current may be unevenly distributed, especially at avalanche breakdown. This can in

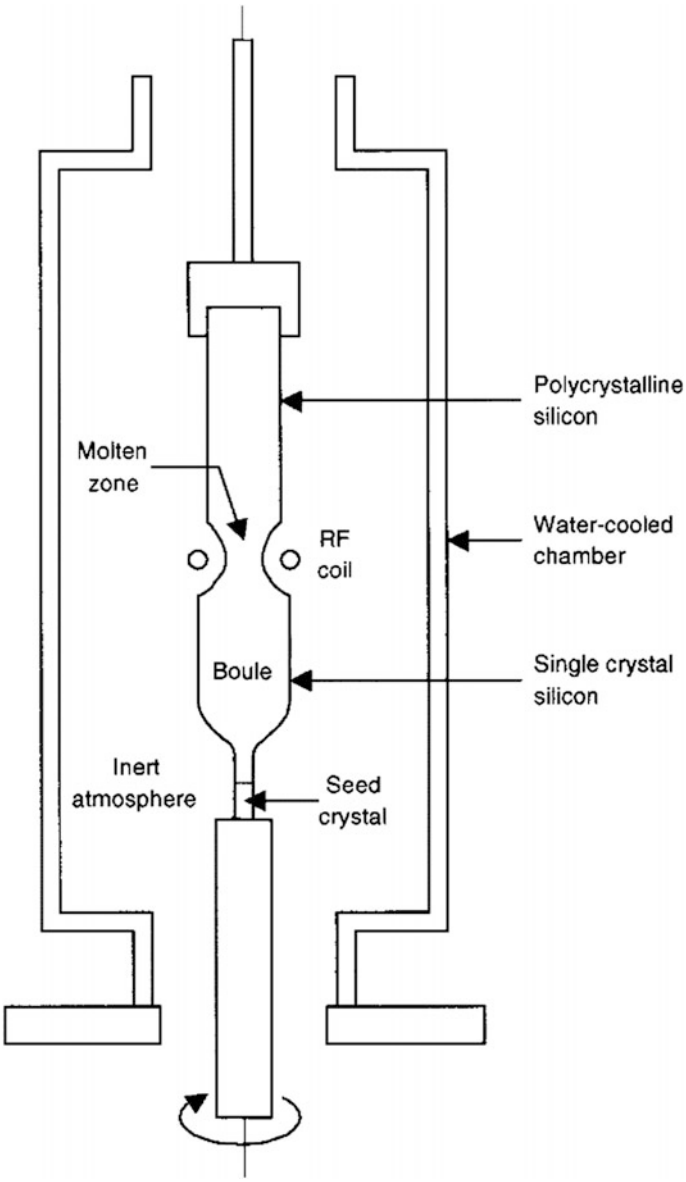


Fig. 4.2 Float-Zone (FZ) process for crystal growth. Figure taken from [Ben99]. Reprint with permission of John Wiley & Sons, Inc

turn result in local overheating and device failure. However, even with the FZ process, which alone is suited for the preparation of the extremely pure silicon crystals needed for power devices, fluctuations in doping cannot be avoided. The obtained wafers

show periodic concentric rings with doping variations called "striations". An example for the resistivity profile measured across a wafer is shown in Fig. 4.3

According to Eq. (2.34), the resistivity variations correspond to doping fluctuations in an inverse manner. Since the doping directly determines the blocking voltage, it is not possible to produce high voltage power devices with the required narrow design, if the doping of the starting material is fluctuating in the amount as shown in Fig. 4.3a. Only with the method of neutron doping (Fig. 4.3b), it became possible to realize for instance thyristors with a blocking voltage of > 2000 V.

The neutron transmutation doping (NTD) of silicon was proposed and investigated first by Lark-Horovitz [Lar51] and Tannenbaum and Mills [Tan59, Tan61]. For the fabrication of 5 kV-thyristors with an acceptable recovery time, the method was reinvented and introduced on industrial scale in 1973/74 [Haa76, Jan76, Scl74]. It is based on the radioactive transmutation of the silicon isotope $^{30}_{14}$Si to phosphorous $^{31}_{15}$P by thermal neutrons. This isotope is contained in natural silicon to 3.09%, besides $^{29}_{14}$Si with a share of 4.67% and the main isotope $^{28}_{14}$Si with 92.23%. As starting material a crystal bar is used which is grown by the FZ method and doped very low, typically below 5×10^{12} cm^{-3}. If the silicon rod is placed near the core of a nuclear reactor, the $^{30}_{14}$Si atoms capture partly a thermal neutron and turn under emission of a γ quant into the unstable isotope $^{31}_{14}$Si which by emission of an electron (β particle) decays then into the stable phosphorous isotope $^{31}_{15}$P:

$$^{30}_{14}\mathrm{Si}(\mathrm{n},\gamma) \rightarrow {}^{31}_{14}\mathrm{Si} \rightarrow {}^{31}_{15}\mathrm{P} + \beta$$

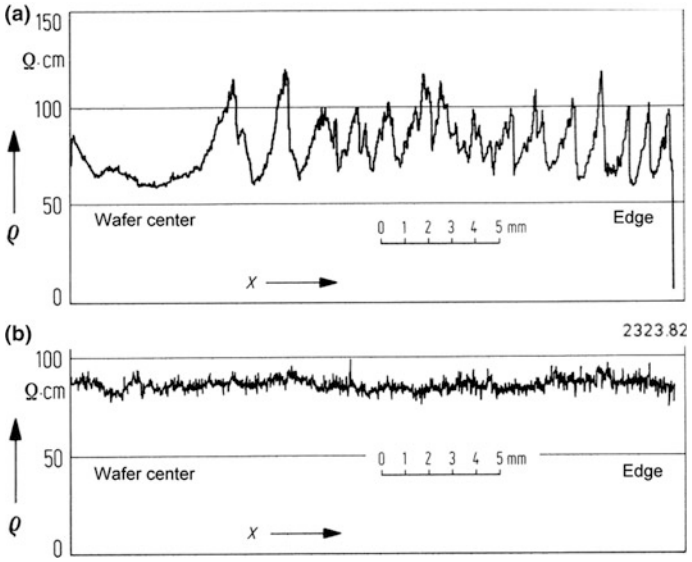


Fig. 4.3 Radial resistivity profile achieved by conventional doping (top) and by neutron transmutation doping (bottom). Figure from [Sco82]

The decay of ^{31}Si to ^{31}P has a half-time of 2.6 h. Thermal neutrons exhibit a decay length in Si of 19 cm, while the diameter of the bars is usually 10 to 20 cm. By rotation of the bar during the irradiation, however, the mean neutron flux in the Si bar becomes nearly homogeneous, the difference between the value at the axis to that at the periphery being less than 4% even for a rod diameter of 15 cm [Jan76]. The final phosphorous concentration obtained by these processes is given by

$$N_{Phos} = c \cdot \Phi \cdot t \tag{4.1}$$

where Φ denotes the neutron flux in $cm^{-2}s^{-1}$ and t the time. With the cross section data given in [Jan76], the constant c for natural silicon is obtained to be $c = 2.0 \times 10^{-4}$ cm^{-1}. The typical flux density Φ of thermal neutrons at the used positions is in the range 10^{13}–10^{14} $cm^{-2}s^{-1}$ [Amm92]. At a flux of 2×10^{13} $cm^{-2}s^{-1}$, for example, an irradiation time of 3.5 h is necessary according to (4.1) to obtain a phosphorous doping of 5×10^{13} cm^{-3}. The phosphorous concentration can be adjusted by this method very exactly within limits of only ±3%.

After the neutron irradiation a storage period is necessary for the on-going transmutation of $^{31}_{14}$Si into $^{31}_{15}$P and particularly for the decay of the activity due to the β emission. The time for the decay of the activity to an undetectable level is 3–5 days. As secondary process, the reaction of the generated $^{31}_{15}$P atoms with the neutrons and the decay of the resulting $^{32}_{15}$P according to

$$^{31}_{15}\mathrm{P}(\mathrm{n},\gamma) \rightarrow {}^{32}_{15}\mathrm{P} \rightarrow {}^{32}_{16}\mathrm{S} + \beta$$

has to be considered. The decay of $^{32}_{15}$P into $^{32}_{16}$S has a half time of 14.3 days and will prolong the time required for the activity decay if the phosphorous concentration generated is higher than about 1×10^{15} cm^{-3} [Jan76]. To eliminate lattice defects induced by the radiation, the crystal bar is annealed at a temperature of 800 °C, before it is further processed to wafers.

Because the neutron doping method is rather costly there have been major efforts of silicon wafer suppliers to improve the FZ crystal growth process, and essential advances have been reached in the last 20 years. The doping tolerances of wafers with a diameter of 200 mm are decreased now to ±12% [SIL06]. Nevertheless, the NTD method is still indispensable for high voltage devices.

4.3 Epitaxial Growth

If a Si device requires a lowly-doped layer with a thickness below 50–100 µm, total wafer is thin and the wafer handling is challenging, especially regarding the breakage rate in the production process, and also the wafer must have a low bow to realize fine pattern structures with photolithography. Therefore, epitaxial growth of a thin n^- layer was usual. In 1997, however, so so-called "thin-wafer technology" was introduced [Las97] for a wafer thickness with 100 µm and below. Even 40 µm

thin-wafer technology was introduced for 400V IGBTs and diodes [Boe11]. The thin-wafer technology competes successfully with epitaxial growth. For unipolar devices with n-layers as thin as possible, epitaxial growth is still standard.

Epitaxial growth is an alternative method to produce high purity single crystalline layers. As a starting material or substrate for the epitaxial growth process, a single side polished CZ-wafer is used. On this wafer a layer with higher purity and higher crystal quality is then grown in the same crystal orientation as the substrate, see Fig. 4.4. The epitaxy process is done significantly below the melting temperature of the semiconductor material.

The growth of epitaxial layers from silicon is made in an enclosed chamber, a reactor, using a vapor phase process. There are different possible processes which use one of the following chemical reactions [Ben99]:

$$\begin{aligned} SiCl_4 + 2H_2 &\rightarrow Si + 4HCl \quad \text{at } 1150 - 1220\,^{\circ}C \\ SiHCl_3 + H_2 &\rightarrow Si + 3HCl \quad \text{at } 1100 - 1175\,^{\circ}C \\ SiH_2Cl_2 &\rightarrow Si + 2HCl \quad \text{at } 1025 - 1100\,^{\circ}C \\ SiH_4 &\rightarrow Si + 2H_2 \quad \text{at } 1150 - 1220\,^{\circ}C \end{aligned}$$

Prior to the growth an intensive mechanical and chemical cleaning of the wafer surfaces is done. The substrates are also cleaned and etched in the process reactor at a temperature of 1140–1240 °C with HCl. The growth is executed in a H_2-atmosphere. For doping, PH_3 (phosphine) or B_2H_6 (diborane) for n-type or p-type, respectively, are added to the hydrogen gas in controlled proportions.

Epitaxial layers in silicon can be fabricated with high purity, especially with very low impurity content of carbon and oxygen.

For the production of SiC monocrystals a high effort is necessary. The SiC substrate crystal growth is done at a temperature of 2300 °C. It was very difficult to achieve monocrystals of low defect density, however there is strong progress. Nitrogen-doped substrates as 150 mm wafer with resistivity of 15–28 mΩ cm are available. Former SiC substrates suffered of a high number of crystal defects, especially dislocations. Screw dislocation propagating through the bulk at the

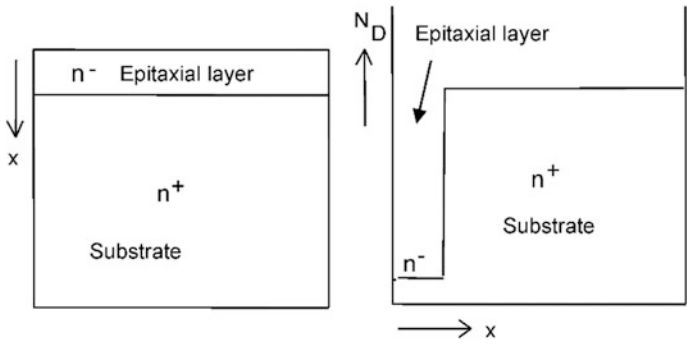


Fig. 4.4 Epitaxially grown wafer in cross section. Example n^+-substrate with n^- epitaxial layer

crystal growth formed micropipes which continue in subsequent epitaxial layers. The former high micropipe density is now significantly improved and micropipe-free wafers are found. The specification at manufacturers is meanwhile < 1 micropipe/cm^2. 6-inch wafers are available, the possibility of 8-inch wafer diameter was demonstrated.

For SiC power devices, a following epitaxy process is always necessary to reach the desired material quality. This epitaxial process is done at 1400–1600 °C under H_2-atmosphere. Commercial SiC CVD processes typically use silane SiH_4 and light hydrocarbons, such as propane or ethylene, and hydrogen as a carrier gas. Growth rates are usually 6–7 µm/h [Cho11]. Using $SiCl_2$ as primary growth species, growth rates of 30–100 µm/h have been obtained, 5–15 times higher than most conventional growths. The deposition temperature was 1750 °C, a good crystal quality and minority carrier lifetime in the order of 0.75 µs were achieved [Cho11]. Also a carrier lifetime in the order of 1–2 µs is reported. Basal plane dislocation (BPD) densities are an obstacle for device fabrication, they are today in the order of 1–2 cm^{-2} and limit the yield for large area devices. For bipolar devices, BPDs are critical defects, since they replicate from the substrate into the epitaxial layer. During bipolar operation the BPDs expand to form stacking faults which degrades the bipolar current capability (bipolar degradation). Much work is done to reduce the BPD density [ECP16]. However, the current methods used to reduce BPDs and to increase lifetime are an high effort, and particularly large area SiC wafers of high quality are therefore very expensive compared to Si.

Due to the lack of GaN large-size monocrystals, GaN is grown by epitaxy on other substrates – Si, SiC and sapphire. Sapphire is an insulator with limited thermal conductivity and only used for microwave devices, not for power devices. Due to cost reasons, Si substrates are preferred, even if devices from SiC substrates show better characteristics [Hil15].

Heterostructures are grown by metal organic chemical vapor deposition (MOCVD) and plasma induced molecular beam epitaxy (PIMBE). In the MOCVD process Triethylgallium $Ga(C_2H_5)_3$ and Trimethylalluminium $(CH_3)_3Al$ are used as source and ammonia as precursor [Amb99]. With moderate growth temperature of 740–780 °C and high ammonia flux, wafers with GaN on Si (111) substrates can be grown with high resistivity [Tan10]. By the share of Triethylgallium and Trimethylalluminium, pure GaN, mixed composition $Al_xGa_{1-x}N$ ore pure AlN can be produced, where x denotes the share of Al in relation to GaN. For more details, see Sect. 4.10.

4.4 Diffusion

The most important way to create n- and p-layers of defined thickness is by diffusion of the impurities into the solid at high temperature. This method to control conductivity was introduced in the early 1950s, and since then various techniques have been explored to improve uniformity and reproducibility. To a high degree,

these requirements can be met carrying out the diffusion with an impurity layer deposited before by ion implantation (see Sect. 4.5). Providing the wafer with a masking oxide pattern (see Sect. 4.6), the diffusion can be restricted to windows in the layer. Due to these opportunities the diffusion is a key process in manufacturing of devices.

For diffusion, semiconductor wafers are placed in a furnace, where they are brought in contact with the dopant, if not only a drive-in of pre-deposited impurities is intended. Often an inert gas which contains the impurity as gaseous molecules, e.g. BCl_3 or PH_3, flows through a tube with the wafers, at which surface the doping atoms are released in a reaction. However, also other techniques are in use [Mue93]. The temperature used for dopants like B and P in silicon ranges from about 1000–1250 °C. If the surface is covered with a masking pattern, the diffusion at the edges of a window proceeds not only normally to the surface but also laterally under the oxide to a certain extent; it is a two-dimensional or, near a corner, a three-dimensional process. Since both the vertical and lateral diffusion are decisive for the design of devices, a precise knowledge of impurity distribution in dependence of diffusion conditions is a necessity. On atomic scale, diffusion in crystals is a complicated process involving crystal defects such as vacancies (vacant lattice sites) and interstitial atoms located between lattice points. Nevertheless the macroscopic description is normally the same as for diffusion in gases and fluids. The differential equation of diffusion is a special case of the equations for electrons and holes developed in Sect. 2.9. In the next subsection, we derive the diffusion equation however separately, before using it to calculate some basic and practically important diffused impurity distributions.

4.4.1 *Diffusion Theory, Impurity Distributions*

If the impurity concentration N is inhomogeneous, a diffusion current with density

$$\vec{J} = -D \cdot gradN(x, y, z, t) \tag{4.2}$$

results, as described already in Sect. 2.6.3. D is the diffusion constant or 'diffusivity' which for most atoms in silicon becomes appreciable only above 800 °C. If also $\vec{J}$ varies in space, this leads in turn to a variation of N with time, as is illustrated for one dimension in Fig. 4.5: If the current density J_2 flowing out of a volume element of thickness dx is higher than the current J_1 flowing in, i.e. if $\partial J/\partial x > 0$, then the particle concentration in the volume element decreases with time (Fig. 4.5b). Here we use that, different from electrons and holes, impurities cannot be created and not disappear (zero generation/recombination). Hence the net flow out of the volume element, $\partial J/\partial x$, yields directly the decay rate of particle concentration: $-\partial N/\partial t = \partial J/\partial x$ or in general form:

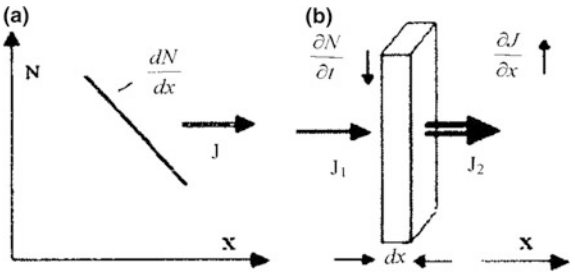


Fig. 4.5 Illustration of the relation between the **a** gradient of the particle concentration and particle current density (first Fick's law) and **b** divergence of particle current density and change of particle concentration with time (continuity equation, second Fick's law)

$$-\frac{\partial N}{\partial t} = div\,\vec{J} \tag{4.3}$$

Inserting (4.2) one obtains the 'diffusion equation':

$$\frac{\partial N}{\partial t} = div(D \cdot gradN) \tag{4.4}$$

Equations (4.2) and (4.4) are called Fick's first and second law, respectively, since formulated first by A. Fick in 1855. Together with the initial and boundary conditions Eq. (4.4) describes how a spatial doping distribution develops with time. For not too high concentrations, the diffusivity D is independent of N and the space coordinates. Then Eq. (4.4) takes the form

$$\frac{\partial N}{\partial t} = D \cdot \Delta N \tag{4.5}$$

with the Laplace operator $\Delta = \partial^2/\partial x^2 + \partial^2/\partial y^2$, if two dimensions are involved. The impurity level up to which this applies depends on the intrinsic concentration n_i at the diffusion temperature, which at 1100 °C for example is about 3×10^{19} cm^{-3} (see Sect. 4.4.3). Most analytical calculations, including the following, deal with the solution of Eq. (4.5).

We consider now a few important cases of diffusion into a half-space $x \geq 0$.

(a) Drive-in diffusion of a thin impurity layer deposited in the semiconductor near the surface: Diffusion processes of this kind are often used in device fabrication. The impurity layer can be introduced by ion implantation or a short prior diffusion. The layer is assumed to be infinitesimal thin and located at $x = 0$, described by a δ-function with integral value S, the deposition dose per area. The solution of the one-dimensional form of (4.5)

$$\frac{\partial N}{\partial t} = D \cdot \frac{\partial^2 N}{\partial x^2} \tag{4.6}$$

must satisfy the initial condition N(x, 0) = 0 for $x > 0$ and the condition

$$\int_0^\infty N(x,t)dx = S = \text{const} \tag{4.7}$$

for all times $t \geq 0$. Equation (4.7) follows, since after the prior deposition no further impurities are introduced and hence their total number remains constant, assuming that out-diffusion into the atmosphere does not occur. The solution of (4.6) which satisfies these conditions is

$$N(x,t) = \frac{S}{\sqrt{\pi \cdot D \cdot t}} \cdot \exp\left(-\frac{x^2}{4Dt}\right) \tag{4.8}$$

As function of x, (4.8) represents a so-called Gaussian distribution. By integration one can verify the condition (4.7). The decrease of the surface concentration $N_0 = S/\sqrt{\pi Dt}$ with time is compensated by the increase of the depth x at which the normalized distribution $N(x, t)/N_0$ takes a given value. With the diffusion length

$$L = 2 \cdot \sqrt{D \cdot t} \tag{4.9}$$

equation (4.8) takes the form

$$N(x,t) = \frac{2}{\sqrt{\pi}}\frac{S}{L}\exp\left(-\left(\frac{x}{L}\right)^2\right) \tag{4.10}$$

A diffusion profile of this type is shown in the earlier Fig. 3.6. From (4.10) the deposition dose and diffusion length needed for the surface concentration and junction depth of that figure can be easily determined. How the distribution (4.10) develops with time is shown in Fig. 4.6 by the solid curves. The x-value at which a certain concentration $N_B < N_0$ is reached, is identical with the junction depth x_j of a pn-junction which is formed if the wafer possesses a background doping of opposite polarity with density N_B. As is shown by the figure, x_j increases somewhat weaker with L than proportional. From (4.10) one obtains

$$x_j = L \cdot \sqrt{\ln\left(\frac{2S}{\sqrt{\pi} \cdot L \cdot N_B}\right)} = L \cdot \sqrt{\ln\left(\frac{N_s}{N_B}\right)} \tag{4.11}$$

Within a limited range of L, the root expression can be considered approximately constant. For the range $10^3 < 2S/(\sqrt{\pi} \cdot L \cdot N_{vol}) < 10^5$ one obtains as a rule of thumb $x_j \approx 6 \cdot \sqrt{D \cdot t}$.

In reality, the thickness Δx of the pre-deposition layer is of course not zero, but may be for example 200 or 600 nm. In the surface region, the solution (4.7) is then valid only for times with $L_D \gg \Delta x$, i.e. $t \gg \Delta x^2/(4D)$. For impurity layers

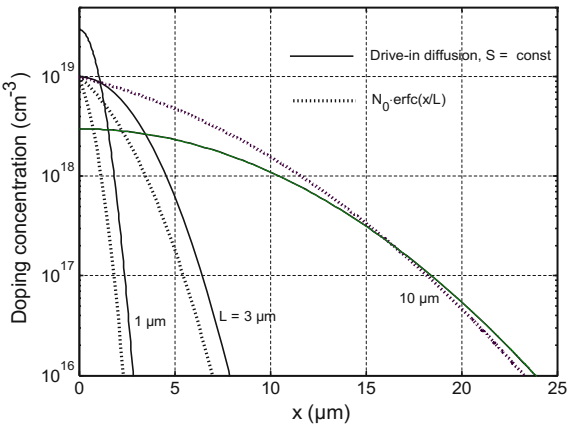


Fig. 4.6 Doping profiles for diffusion with constant integral number $S = 2.66 \times 10^{15}$ cm^{-2} per area (solid curves) and constant surface density (dotted curves). Assuming a diffusion constant $D = 1 \times 10^{-12}$ cm^2/s = 0.36 µm^2/h (for P and B at 1200 °C, see Fig. 4.11), the three L-values correspond to diffusion times t = 0.694, 6.25 and 69.4 h, respectively

produced by ion implantation, however, Eq. (4.10) can be modified in a simple manner to include the distribution in the layer at small t (see Sect. 4.5).

(b) Diffusion with constant surface concentration: A constant impurity concentration at the semiconductor surface during the diffusion process can be achieved by supplying the dopant as admixture to an inert gas flowing around the wafers or from vapor phase of the dopant. Another possibility is to deposit a solid layer with higher dopant concentration than its solubility in the semiconductor. The surface density is then maintained at the solubility value. The solution of (4.6) which satisfies the boundary condition $N(0,t) = N_s = \text{const}$ for $t > 0$ and the initial condition $N(x,0) = 0$ for $x > 0$ is given by

$$N(x,t) = N_s \cdot erfc\left(\frac{x}{2\sqrt{D\,t}}\right) \tag{4.12}$$

with the complementary error-function defined as[1]

$$erfc(x) = 1 - \frac{2}{\sqrt{\pi}} \int_x^x e^{-x'^2} dx' \tag{4.13}$$

[1]This function and the normal error function erf(x) = 1 – erfc(x) appear also in other diffusion processes (see next section, case c). Often the following analytical approximation for x $\geq$ 0 is sufficient:

$$erfc(x) \approx \exp(-1.14x - 0.7092x^{2.122})$$

Its maximum error in the range 10^{-7} < erfc < 1 is 2‰. For negative arguments appearing in Eq. (4.16) below the approximation can be used considering that erf(–x) = –erf(x) = erfc(x) – 1.

As is shown by (4.12), the doping density is only a function of x/L in this case. Profiles given by (4.12) are shown in Fig. 4.6 as dotted curves, using the same L-values as for the curves with constant number S of impurities discussed in case a). For equal surface concentration and L-value, the *erfc*-profile penetrates less deep into the crystal than the Gaussian distribution, as shown for $L = 3$ µm. This may be surprising at first sight, but follows because the concentration in the case of drive-in develops from higher values at smaller times, as can be seen from the curves for $L = 1$ µm. Later, for $L = 10$ µm, the *smaller* surface concentration of the drive-in distribution leads to *reduced* penetration depth. To get deep-reaching diffusion profiles with simultaneously high surface density, a diffusion with constant surface density is advantageous, since otherwise a very high deposition dose S would be necessary.

The integral impurity number per area increases now with time. Integrating (4.11) over x one obtains:

$$S = \int_0^\infty N(x,t)dx = N_s \cdot 2\sqrt{Dt}/\sqrt{\pi} \tag{4.14}$$

since $\int_0^\infty erfc(x)dx = 1/\sqrt{\pi}$. If a short diffusion with constant surface concentration is used for pre-deposition, Eq. (4.14) yields the deposition dose to be used in (4.8) for drive-in.

The diffusion case with constant surface concentration is used besides others for comparison of theory and measurements. It is found, that the experimental profiles of the group III and V dopants are in agreement with Eq. (4.12), as long as the surface concentration is below about 1×10^{19} cm^{-3}.

(c) Drive-in diffusion with a deposition layer of limited width: In this case the diffusion proceeds not only vertically to the surface but near the edges of the window also sideways or laterally under the mask (Fig. 4.7). The lateral diffusion is generally not undesirable but on the contrary widely used in device concepts. A well-known fact is that the lateral penetration depth under the oxide, y_{pd}, is always smaller than the vertical penetration depth x_{pd} in the central region of the window. In most cases, the relation y_{pd}/x_{pd} is in the range $\approx 0.6 - 0.9$. The two-dimensional impurity distribution is described often in a simplified manner, using for the region below the window, $-d \leq y \leq d$, $x \geq 0$, the unchanged one-dimensional profile obtained without mask, whereas the lateral decay is modeled by a cylindrical distribution under the oxide. An exact mathematical analysis of diffusion through a window

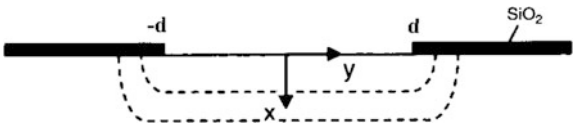


Fig. 4.7 Vertical and lateral diffusion through a window in a masking layer

limited on one side by a mask has been given both for the case of constant surface concentration as well as constant amount of impurities by Kennedy and O'Brien [Ken65]. In the following, the latter case, which is the more important and simultaneously simpler, is examined for a strip-like window.

As in case (a), the initial impurity layer is assumed to be infinitesimally thin and located at the surface $x = 0$, now however only in the stripe $-d \leq y \leq d$, $-\infty < z < \infty$. The integral of the $\delta(x)$-function representing the initial distribution is given by the doping dose S per area. Outside the window the concentration is everywhere zero at $t = 0$. Since no impurities are introduced later and losses by out-diffusion are excluded here, one has for all $t \geq 0$:

$$\int_{x=0}^{\infty} \int_{y=-\infty}^{\infty} N(x', y', t \geq 0) dx' dy' = 2d \cdot S = \text{const} \tag{4.15}$$

The solution of the differential Eq. (4.5) which satisfies these conditions is given by:

$$N(x, y, t) = \frac{S}{\sqrt{\pi} L} \cdot e^{-\left(\frac{x}{L}\right)^2} \cdot \left\{ erf\left(\frac{y+d}{L}\right) + erf\left(\frac{d-y}{L}\right) \right\}, \tag{4.16}$$

where as before $L = 2\sqrt{Dt}$ and the coordinate system of Fig. 4.7 is used. *erf* denotes the normal error-function defined as

$$erf(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-x'^2} dx' = 1 - erfc(x) \tag{4.17}$$

The distribution (4.16) differs from (4.10) by the factor

$$F(y) = \frac{1}{2} \cdot \left(erf\left(\frac{y+d}{L}\right) + erf\left(\frac{d-y}{L}\right) \right) \tag{4.18}$$

which describes the lateral variation of N. $F(y)$ satisfies the one-dimensional diffusion equation in y as follows from the above case (b); similarly, the remaining factor in (4.16) was found under (a) to be a solution of (4.6). From this it follows in a simple manner that the product solves the two-dimensional Eq. (4.5). The integral of $F(y)$ from $-d$ to d is by the same amount, namely $L \int_0^{2d/L} erfc(u) du$, smaller than $2d$ as is contributed by the integral over the outer regions, so also the condition (4.15) is satisfied.

According to Eq. (4.16), $N(x, y)$ obeys at all points y the same Gaussian x-dependency (in relative units), including the region under the oxide. Likewise the y-dependency expressed by $F(y)$ is the same for all distances x from the surface. In Fig. 4.8, $F(y)$ is plotted for several diffusion lengths L assuming a window width

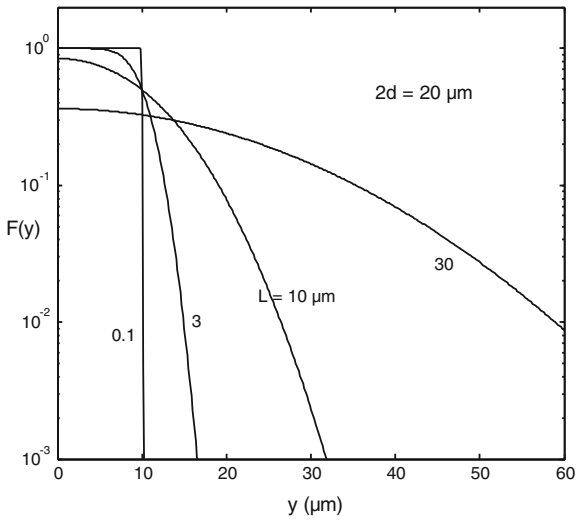


Fig. 4.8 Lateral distribution function $F(y)$ for various diffusion lengths L. Window width $2d = w = 20$ µm

$2d = w = 20$ µm. For $L = 0.1$ µm, $F(y)$ is box-shaped, meaning that there is virtually still no lateral diffusion. With increasing L this changes. As long as $w/L > 4$, the value of $F(y)$ in the center of the window at $y = 0$ is equal to 1 whereas up to the edges it decreases to ½. The above mentioned assumption of y–independent concentration in the window region hence fails considerably. For $y > d$, the concentration decreases as $0.5 \cdot erfc((y - d)/L)$ for $w/L > 2$, similar for $y < -d$. For long diffusion times, cases $L = 10$ and 30 µm in Fig. 4.8, also the concentration in the central region is reduced by the lateral diffusion. The reduction of the impurity amount in the window region by the lateral diffusion is a point to be considered selecting the diffusion conditions.

Curves of constant concentration N of the function (4.16) are shown for an example in Fig. 4.9. The window width is in a range where the mask at one side has no influence on the impurity distribution at the other side for the used diffusion length. The lateral decrease of the concentration within the window from the maximum value $N_0 = 2 \cdot S/(\sqrt{\pi} \cdot L) = 3.76 \times 10^{18}$ cm^{-3} is conspicuous. As can be seen, furthermore, the lateral penetration depth y_{pd}, defined as the distance from the edge of the window to the point of a given concentration $N < N_0/2$ at the surface is smaller than the vertical penetration depth x_{pd}, i.e. the vertical distance to this concentration deep in the window. The penetration depths coincide with the lateral and vertical junction depths $y_j(N_B)$, $x_j(N_B)$, if a background doping of opposite polarity with density $N_B = N$ is present.

Equation (4.16) involves an interrelation between x_{pd} and y_{pd}: With the origin of the y-coordinate at the window edge as in Fig. 4.9, the concentration at $x = 0$, $y = y_{pd}(N)$ is per definition equal to the concentration at $x = x_{pd}(N)$, $y < -2L$. Hence

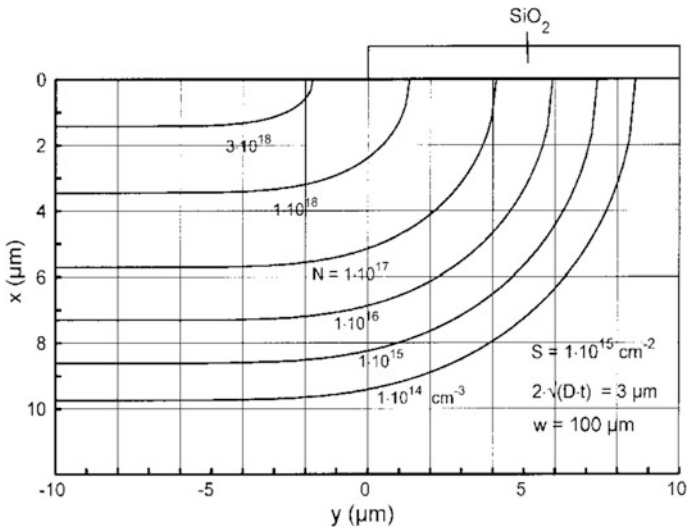


Fig. 4.9 Curves of constant concentration for a drive-in diffusion near the diffusion mask, according to Eq. (4.16). Deposition dose 1×10^{15} cm^{-2}, window width 100 µm, $L = 3$ µm

$$\frac{N}{N_0} = \frac{1}{2} erfc(y_{pd}/L) = \exp\left(-(x_{pd}/L)^2\right) \tag{4.19}$$

where $w/L > 4$ is assumed so that the window width drops out. Solving for x_{pd}/L one obtains for the ratio of penetration depths:

$$\frac{y_{pd}}{x_{pd}} = \frac{y_{pd}}{L} \Big/ \sqrt{\ln\left(\frac{2}{erfc\left(y_{pd}/L\right)}\right)}, \tag{4.20}$$

According to (4.20), the ratio y_{pd}/x_{pd} increases with increasing y_{pd}/L and hence with decreasing N/N_0. In Fig. 4.10, y_{pd}/x_{pd} is shown as function of N/N_0 using Eq. (4.19) to correlate y_{pd} with N. The variation of y_{pd}/x_{pd} is in accordance with the above-mentioned experimental observations.

In MOSFETs and IGBTs, the n-channel is formed over the rim of the p-base stemming from lateral diffusion. Equation (4.16) provides a relative simple tool to determine the diffusion conditions required for the appropriate channel length and doping concentration under the gate. In cases where the neglected loss of dopant by out-diffusion or possibly by oxide growth is significant or if the variation of the diffusivity with N is considerable, more complex numerical process simulation is needed.

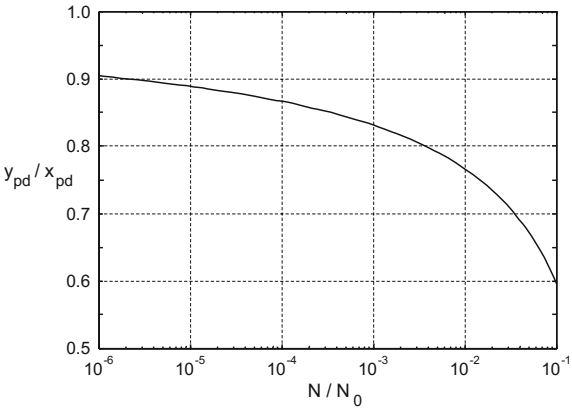


Fig. 4.10 Ratio of lateral to vertical penetration depth as function of the defining concentration N divided by the concentration N_0 in the middle of the window

4.4.2 *Diffusion Constants and Solubility of Dopants*

The diffusion of impurities is thermally activated, hence the temperature dependence of the diffusion constant is given by an Arrhenius relationship

$$D = D_0 \cdot e^{-E_A/kT} \tag{4.21}$$

where E_A is the activation energy and T the absolute temperature. Both E_A and the pre-factor D_0 can be considered as constant.

For the commonly used dopants in silicon the parameters in (4.21) are compiled in Table 4.1. The values are representative and apply to typical diffusion processes for power devices. Diffusion parameters in the literature differ partly considerably from one another, depending on the conditions of measurements. The activation energies fall in the range 3.3–4.2 eV. A plot of the diffusivities versus temperature calculated with these parameters is shown in Fig. 4.11. For a temperature rise of 100 °C, the diffusion constants increase approximately by an order of magnitude.

Aluminum is the fastest diffusing dopant, its diffusion constant is about two orders of magnitude higher than that of the slowest doping elements, As and Sb, and about a factor 20 higher than that of P and B, which are most commonly used for diffusion. P and B as well as also As and Sb have nearly equal diffusivity.

Table 4.1 Parameters of diffusion constants in the Arrhenius relationship (4.21)

Element	D_0 [cm^2/s]	E_A [eV]	Reference
B	0.76	3.46	[Sze88]
Al	4.73	3.35	[Kra02]
Ga	3.6	3.5	[Ful56]
P	3.85	3.66	[Sze88]
As	8.85	3.971	[Pic04]
Sb	40.9	4.158	[Pic04]

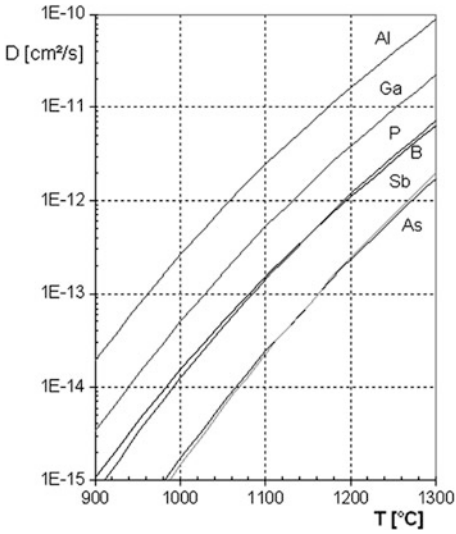


Fig. 4.11 Diffusion constants of dopants in silicon as a function of temperature

The usability of substances for diffusion (and generally doping) can be judged only if also the solubility in the semiconductor is taken into account. Figure 4.12 shows the solubility of impurities in silicon in dependence of temperature. Besides the normal dopants, also oxygen, Au and Pt are included. All elements are only little solvable. Related to the density of Si atoms, 5.0×10^{22} atoms/cm^3, even the highest solvable impurities, phosphorous and arsenic, have a solubility smaller than 1%. The solubilities of P and As given in the figure [Bor87] are a factor 3–4 smaller than often cited previous values [Tru60]. The maximum carrier concentration obtainable with a doping element approaches to a high share the new solubility limits, meaning that nearly all the doping atoms are electrically active. The solubility of electrically inactive dopants on interstitial sites or in complexes of more than one atom is only small. Apart from a region near the melting point of Si the solubilities decrease with decreasing temperature. This can lead to precipitation if the system is slowly cooled down.

Whereas Al has the highest diffusivity, its solubility is the smallest of group III and V elements. Al is used for creating deep pn-junctions especially in thyristors in which pn-junctions with a depth up to 100 µm are used. Its solubility, however, is not sufficient for a surface concentration necessary for good ohmic contacts. Therefore, when used for deep diffusions, a second and for some thyristors even a third p-type diffusion step is carried out: The Al diffusion is followed by a Ga diffusion, and since also Ga has, next to a good diffusivity, only a moderate solubility, some manufacturers use a third diffusion step with boron. In Fig. 8.3, a diffusion profile of a thyristor manufactured in this way is shown. Acceptor profiles with a penetration depth in the range of 20 µm are made preferably with boron. Besides its high solubility it can be masked by SiO_2.

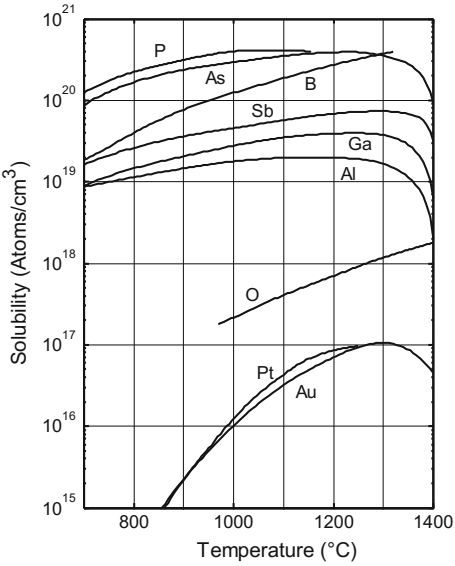


Fig. 4.12 Solubility of some impurity elements in silicon. For P, As, B, Sb the data are from [Bor87], for Ga, Al and O from [Tru60], for Au and Pt from [Bul66] and [Lis75], respectively

Donor profiles for power devices are almost solely generated with phosphorous. P is the only n-doping element in silicon, which has a sufficient solubility and acceptable diffusion constant. Even with phosphorous, deep n-profiles, needed for example for bipolar transistors, require a high diffusion temperature and a long diffusion time. For the 120 µm deep profile of the collector layer of a npn-bipolar transistor (Fig. 7.3), a diffusion time of about 140 h at a temperature of 1260 °C is necessary.

Because of its small diffusivity, arsenic is not used for diffusion in manufacturing of power devices, which contrasts to VLSI technology, where As is widely used just because of its small diffusivity. Because of its high *solubility*, arsenic is used in power devices as dopant of substrate wafers for epitaxy. With As, a very low specific resistivity of less than 5 mΩ cm can be reached while only values near 15 mΩ cm are possible with Sb, the alternative n-type dopant of substrates. Since in many power devices the resistivity of the substrate determines the series resistance, a very low resistivity is important for low conduction losses. If a higher resistivity can be tolerated, substrates are doped with antimony, because unwanted doping of the epitaxial layer by transfer from the substrate (autodoping) can be minimized with Sb.

As mentioned, the given diffusion constants refer to typical diffusion conditions and not too high impurity concentrations. Typically, the diffusion is carried out in an inert ambient. Under oxidizing conditions, the diffusion of B and P is accelerated, and higher diffusion constants are measured [Kar95]. At high doping concentrations the diffusivity increases with N, as is indicated by a flattening of the impurity profile in the surface region, see Fig. 4.13 further below.

4.4.3 High Concentration Effects, Diffusion Mechanisms

At high doping concentrations, there is firstly the **field effect** which leads to an increase of the effective diffusion constant with concentration. Since dopants are ionized, the built-in field associated with a diffused impurity distribution (see Sect. 3.1.2) causes a field component of the impurity particle current. Whereas the field current of *carriers* produced by the built-in field compensates the diffusion current, the opposite charge of the doping ions implicates that the field current of impurities has the same sign as the diffusion current. Analogous to (3.32) the field in a donor region is $E = -kT/q \cdot d\ln n/dx$. Using Eq. (2.20) and the Einstein relationship (2.44) the field current of impurities is obtained as

$$J_E = -D_0 \cdot N \cdot \frac{d\ \ln n}{dx} = -D_0 \cdot N \frac{dN/dx}{\sqrt{N^2 + 4n_i^2}} \tag{4.21}$$

where D_0 is the low concentration diffusion constant. Because of the proportionality to dN/dx, the field current is taken together with the diffusion current expressing the total impurity current in the form (4.2), $J = -D \cdot dN/dx$, with the effective diffusion constant

$$D = \left(1 + \frac{N/2n_i}{\sqrt{1 + (N/2n_i)^2}}\right) \cdot D_0 \tag{4.22}$$

With increasing N, D increases around the intrinsic concentration at the diffusion temperature by a factor 2. Especially in the region $0.2 \leq N/n_i \leq 4.1$, where D varies from 1.1 to 1.9 times D_0, the diffusion equation in the general form (4.4) should be used. From Eqs. (2.6), (2.8), (2.9) together with (2.25), (2.27), the intrinsic concentration at 1200 °C, for example, is obtained to be 4.3×10^{19} cm^{-3} for a doping concentration $N = 1 \times 10^{19}$ cm^{-3}.

However the *observed* increase of D with N is for the dopants in silicon stronger than explainable by the field effect. The cause of this lies in the **diffusion mechanism** on atomic scale. The doping impurities are of the substitutional type, they substitute silicon atoms on lattice sites. Such impurities can diffuse only with the help of crystal defects such as vacancies and interstitials, since a movement directly by exchange with adjacent silicon atoms is energetically very improbable. There are two mechanisms which are made responsible for the observed diffusion constants: the vacancy-assisted and the interstitialcy mechanism using silicon interstitials. Whereas previously the vacancy assistance was assumed to determine the diffusion of dopants in silicon, there is now evidence that this mechanism is used only by Sb and partly As, the slowest diffusing doping atoms, which are larger in size than Si [Ura99]. In the vacancy-mediated diffusion the impurity jumps step by step from one substitutional site to a neighboring one, when there is a vacancy. After each step the probability is high that the impurity jumps back to the vacancy located now

at the old place of the impurity. Since also the vacancy concentration is rather small in silicon [Kar95], the vacancy-assisted diffusion is slow and requires a high temperature.

The elements with higher diffusion constants, P, B, Ga and Al, diffuse by the interstitialcy mechanism [Ura99]. This diffusion mechanism of substitutional impurities is initiated by an interstitial silicon atom (self-interstitial) displacing the impurity from its lattice site and occupying the place itself (kick-out mechanism). The impurity migrates then interstitially to a vacancy or kicks out a silicon atom to take its final or intermediate substitutional site. Although the fraction of interstitial impurities is small, the much faster interstitial diffusion enables this mechanism to produce the higher diffusion constants.

The increase of the diffusivity with N beyond the field effect now is assumed to be due to different charge states of the involved defects, for example as I^-, I^0, I^+ for self-interstitials. The relative concentrations of these states depend on the Fermi level: According to Eq. (2.2) the ratio c^-/c^0 of negative to neutral point defects densities is proportional to $\exp(E_F/kT)$ in thermal equilibrium. Since with increasing impurity density above the intrinsic concentration the Fermi level moves towards the respective band edge (see Fig. 2.10), this results in an increase of the concentration of negatively charged point defects in the case of donor doping and of positive point defects in the case of acceptors, each in relation to the density c^0. Assuming c^0 to be independent of x, the total concentration of involved defects and hence the diffusivity itself increases with doping density [Tsa83]. The assumption of homogeneity of concentration c^0 is justified by the much faster diffusion of vacancies and self-interstitials compared with doping atoms. This explains also the observed constancy of the diffusivity below n_i. Impurities with a *larger* diffusion constant than the involved crystal defects show completely different diffusion profiles, as will be seen for gold and platinum in Sect. 4.9.

How the diffusion profile is changed by the increase of D with N is shown for an example in Fig. 4.13. As reported for phosphorous, the diffusivity is assumed to be proportional to N^2 at high concentrations [Fai81] and on the whole can be expressed as

$$D = D_0 \cdot \left(1 + (N/N_{tr})^2\right)$$

The transition concentration $N_{tr} = 3 \times 10^{19}$ cm^{-3} is chosen to reflect the reported increase of D by an order of magnitude in the range up to $N = 1 \times 10^{20}$ cm^{-3} [Fai81, Sze02]. The value equals approximately the intrinsic concentration at 1100 °C, if bandgap narrowing at $N = N_{tr}$ is taken into account using (2.25), (2.27). Due to the strongly enhanced diffusivity the slope of the profile (solid curve) is reduced in the surface region compared with the *erfc*-distribution for $D = D_0$ (dashed curve). The penetration depth is enhanced. The profile depends only on the single variable $x/\sqrt{t}$, as is the case always for constant surface concentration if the diffusivity depends only on N. Using this variable the one-dimensional form of the

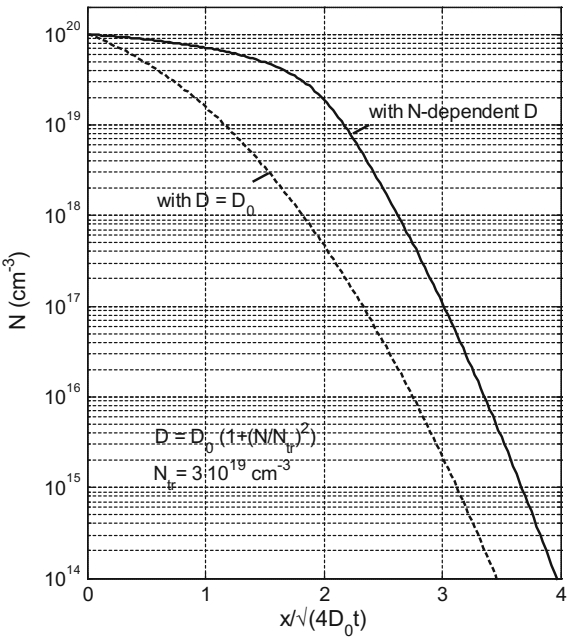


Fig. 4.13 Diffusion profile for a diffusivity increasing quadratically with doping concentration (solid line) compared with the *erfc*(*x*/*L*)-distribution for $D = D_0$ (dashed). The $D(N)$ data refer to phosphorous, see the text

diffusion Eq. (4.4) transforms into an ordinary differential equation [Cra56], which is used to calculate the diffusion profile.

In contrast to silicon, the diffusion process cannot be used to introduce dopants in SiC, because the diffusion constants in SiC are too small. Apart from epitaxial processes, which are less versatile, the only possible method to produce required device structures is ion implantation.

4.5 Ion Implantation

In the ion implantation process atoms of the dopants are ionized and accelerated in an electric field to form a beam of mono-energetic ions. The focused beam is scanned vertically and horizontally and the particles are ejected into the wafer homogenously distributed across the wafer surface. The ions will slow down and stop in the semiconductor. The kinetic energy of the impacting ions is lost to the wafer by two types of collisions; with the cores of the lattice atom – elastic nuclear collisions – and by retardation in the electron shells of the lattice atoms – electronic deceleration. The nuclear collisions will also scatter the ions randomly, giving rise to a distribution of implanted ions where the mean penetration depth is mainly determined by the initial energy of the ions.

The dose of the implanted ions and thereby the quantity of the dopants can be controlled very exactly and it is also possible to mask different areas of the wafer to prevent ions from reaching the semiconductor at these locations. Power semiconductors with challenging technology are often doped using ion implantation. Typically, all semiconductor devices with MOS-structures at the surface, as power MOSFET and IGBT, use ion implantation to form the p- and n-type regions.

The generated profile of the dopants can be described in a simplified way with a Gauss function

$$N(x) = \frac{S}{\sqrt{2\pi} \cdot \Delta R_{pr}} \cdot e^{-\frac{(x-R_{pr})^2}{2 \cdot \Delta R_{pr}^2}} \tag{4.23}$$

where R_{pr}, the projected range, corresponds to the mean depth of the implanted ions, i.e. the peak of the profile in this simplified description. ΔR_{pr}, the projected range straggling, is the statistical variation around this mean value, similar to the standard deviation. The integral amount of doping atoms S corresponds to the implanted dose. A doping profile calculated with Eq. (4.23) is drawn in Fig. 4.14.

The projected range depends primarily on the implantation energy, this relation is shown for boron in Fig. 4.15. Ions that are heavier than boron will have a smaller R_{pr} for the same energy.

With increasing energy, the peak height of the doping profile will decrease due to the increasing straggling. This is shown for boron in Fig. 4.16, where a linear scale is used for the concentration. Most often a log-scale is used, since the concentration varies over many orders of magnitude. Tables for the projected range and projected range straggling for the different ions are found in the literature, e.g. in [Rys86]. The ion distributions can also be simulated using for instance the SRIM software, which is a widely used program available on www.SRIM.org [Zie06].

The description discussed up to now has assumed a solid target with unorganized distribution of atoms, i.e. an amorphous target. But in a single crystalline

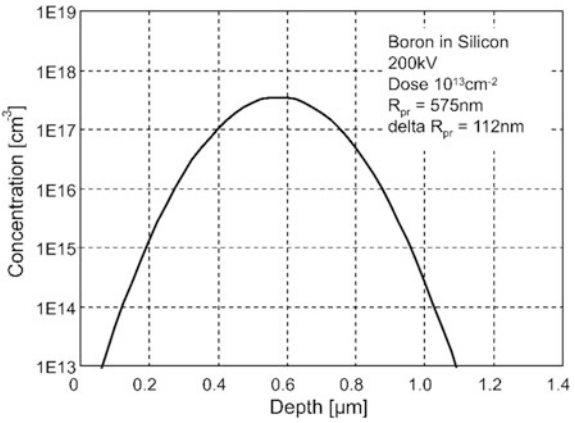


Fig. 4.14 Simplified doping profile of an ion implantation of boron in silicon

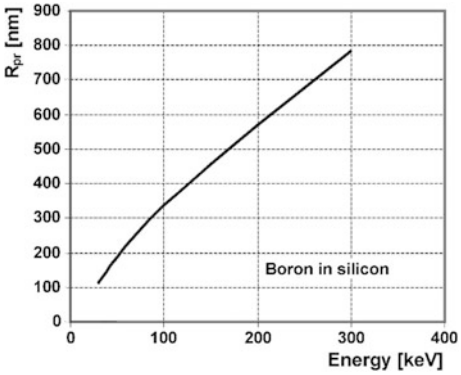


Fig. 4.15 Dependency of projected range R_{pr} on energy for implantation of boron in silicon

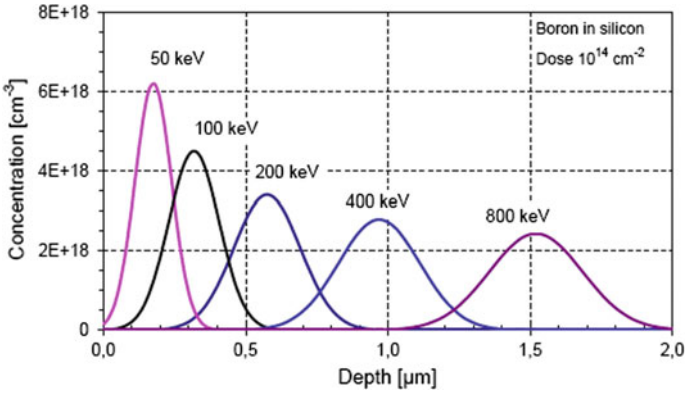


Fig. 4.16 Calculated doping profiles according (4.23) after implantation of boron in silicon with different energies. R_{pr} and ΔR_{pr} are calculated with SRIM [Zie06]

target the atoms are regularly ordered. The elementary cell of the silicon lattice has been already shown in Fig. 2.1. In Fig. 4.17 a drawing of the arrangement of atoms at a view along the [110] direction into a silicon crystal is shown.

If the direction of the penetrating ion beam is aligned with a major crystal orientation, it can penetrate deeper. In the so-called channels, nuclear collisions are very rare and many of the ions are not deflected from their original path. The retardation occurs only due to inelastic impacts with the electron shells of the lattice atoms, i.e. due to the electronic deceleration. The penetration depth in the channels can therefore be 10-times higher than the projected range in an amorphous solid object.

Normally one tries to avoid the channeling, and to counteract this effect the wafer is tilted, so that the implantation occurs in a direction deviating from the major crystal axis. In Fig. 4.18 the dependence of the channeling effect on the tilt angle is shown. Even at a tilt angle of 8° a partial channeling is still visible.

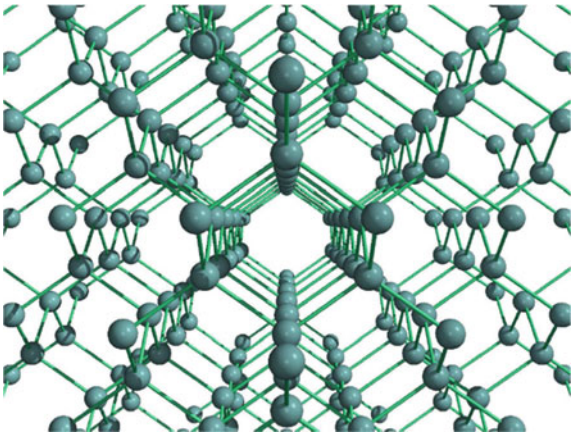


Fig. 4.17 Atomic structure of silicon: viewed in the [110] direction through a silicon crystal of 3 × 3 × 3 unit cells. Figure from [Pic04] © 2004 Springer

A further increase of the tilt angle would cause channeling into other channels. Usually semiconductor wafers are tilted by 7° during ion implantation.

Another way to reduce the channeling is by covering the crystal with an amorphous layer, for instance SiO_2. In this amorphous layer the ions are scattered and the ions will no longer travel in parallel paths. Oxides with a thickness of just 10– 20 nm reduce effectively the channeling effect.

An increased target temperature results in increased amplitude of the lattice oscillations and this will also increase the number of ions that are scattered at the surface. Therefore, the channeling effect is reduced with increased target temperature.

However, none of these countermeasures is sufficient to avoid the channeling effect completely [Rys86]. The best way to reduce the channeling is to use a previously amorphized target. If the silicon wafer is first implanted with Si ions at a high dose, an amorphous Si layer of sufficient thickness is created at the surface, which will effectively prevent the channeling.

When dopants are introduced by ion implantation, one serious complication is the generation of lattice defects by the elastic collisions. These defects consist of atoms knocked out of their lattice positions into interstitial sites, with vacant lattice sites left behind. Combinations of these defects are also known, for instance the Si divacancy, formed by two neighboring vacancies. The maximum amount of ion beam induced lattice defects is located just before the ion comes to rest, e.g. at an implantation of boron in Si at $0.8 \cdot R_{pr}$. The profile of lattice defect is reaching up to the semiconductor surface. The distribution of lattice defects in comparison to the distribution of implanted atoms is shown in Fig. 4.19 on example of an implantation of heavy As ions into silicon.

If the implantation dose of is sufficiently high, an amorphous layer can be generated due to the large number of lattice defects. For every ion there is a critical dose for silicon amorphization, which depends on temperature. The higher the mass of the ion, the lower is the critical dose. However, there is a balance between defect build-up by the incoming ions and the recombination of Si interstitials and

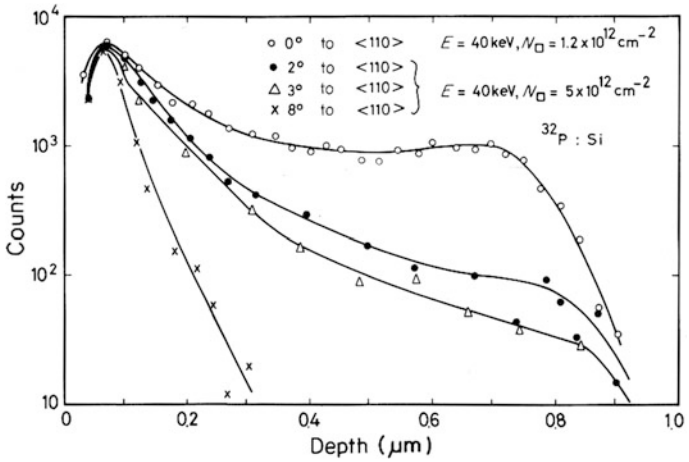


Fig. 4.18 Dependency of the channeling effect on the tilt angle between the ion beam direction and the wafer surface normal for a phosphorus implantation in silicon [Dea68]. Copyright 1968 National Research Council of Canada

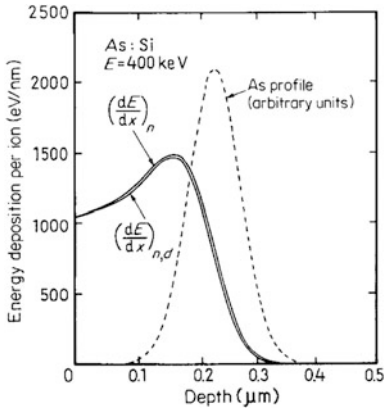


Fig. 4.19 Energy deposition for an implantation of arsenic in silicon in comparison to the distribution of implanted atoms. The profile of lattice defects follows the profile of energy deposition. Figure taken from [Rys86] Original copyright 1971 Phys. Society Japan JPSJ Vol. 31, pp. 1695–1711

vacancies, and this balance depends very much on the target temperature and the flux of incoming ions. For boron implantation at room temperature for instance, no amorphous layer can be generated.

The implantation process is always followed by a thermal annealing process and there are two reasons for it:

- The lattice must be restored to its crystalline form and remaining point defects should recombine.
- The implanted donor- or acceptor ions must be positioned at substitutional sites where they become electrically active, i.e. contribute with an electron to the conduction band or a hole to the valence band.

The annealing of lattice defects starts already at room temperature, but defect complexes of higher order anneal only at a higher temperature. Mostly used in Si processing is a fast annealing process – rapid thermal annealing, RTA, to keep the effect of diffusion of the implanted ions small during annealing. In an RTA process the wafer is heated in a short time to a high temperature by using very intensive light radiators. The temperature is typically above 1000 °C and the annealing time is only some seconds or half a minute. A fast cooling step follows. With this process an effective electrical activation of the doping atoms and an effective annealing of lattice defects is possible without a significant increase of the penetration depth of the doping profile. Even amorphous layers can be restored to monocrystalline layers. Careful optimization of the implantation- and annealing processes is necessary to achieve the desired device structures in the field of micro- and nanoelectronics. For power devices, higher penetration depths than can be reached by conventional implanters are often necessary. In these cases, a diffusion step follows the ion implantation step. However, the process of ion implantation is frequently used because of the possibility of an exact control of the dose and thereby a very exact adjustment of the resulting profile. Modern implanters can produce doping profiles with a deviation of less than 1% across 8-inch wafers.

4.6 Oxidation and Masking

One of the great advantages of silicon compared with other semiconductors is the ability to form an oxide of high quality. Silicon dioxide, SiO_2, is widely used for thin insulating layers (e.g. gate insulation in MOSFETs), and for protecting or masking certain areas of a wafer during processing. SiO_2 features a disordered amorphous structure. For the oxidation of silicon two processes are used.

Dry oxidation:

$$Si + O_2 \rightarrow SiO_2$$

This process gives a low growth rate of the oxide, but the quality is very good. It is used for fabrication of thin oxide layers, which are needed as scattering layer for ion implantation to prevent channeling, and also for the creation of gate oxide in field controlled devices, so-called metal-oxide-semiconductor (MOS) structures.

Wet oxidation:

$$Si + 2H_2O \rightarrow SiO_2 + 2H_2$$

The growth rate of the oxide is higher with this process. It is used to fabricate oxides for masking purposes and oxides as passivation layers. After the wet oxidation step, a dry oxidation step to improve the surface quality of the oxide layer follows in many applications.

The thickness of the oxide layer, d_{ox}, can be calculated with [Ben99]

$$d_{ox} = d_0 + A \cdot t \quad \text{for thin oxide}$$
$$d_{ox} = B \cdot \sqrt{t} \quad \text{for thick oxide.}$$

The constants A and B are temperature dependent. For instance, an oxide layer of the thickness of 1.2 µm for masking purpose is created with a wet oxidation at 1120 °C and a time duration t = 3 h. The process is finished with a dry oxidation in the same reactor for 1 h at the same temperature, which improves the oxide quality.

The diffusion constants of the dopants B, P, As and Sb in SiO_2 are several decades lower than their diffusion constants in crystalline Si. This is used in the manufacturing process. The oxide layer is structured with a photolithographic process. The photoresist is exposed to ultraviolet light through a photomask with the intended pattern. After developing and hard baking of the photoresist, the SiO_2 layer is etched in buffered hydrofluoric acid NH_4F/HF, and finally the photoresist is removed ("stripped"). After removal of the photoresist the wafer has a pattern of SiO_2 layers as shown in Fig. 4.20a.

After cleaning, this pattern can now be used as a mask for a diffusion process. A phosphorus diffusion is typically executed from the vapour phase in a diffusion furnace. With the parameters temperature and time the penetration depth x_j is adjusted. However, the time can only be so long as the phosphorus does not penetrate the masking oxide layer with the thickness d_{ox}. With an oxide thickness of 1.2 µm a penetration depth x_j in the range of 10 µm can be achieved. During diffusion a layer of phosphor silicate glass is growing at the surface of the silicon and of the oxide, it acts as source of diffusion atoms. This layer is subsequently removed.

During the vertical diffusion process a simultaneous lateral diffusion under the oxide takes place. An exact analytical solution was given in Sect. 4.4, see Eqs. (4.15)–(4.18) and Fig. 4.9.

In the mathematical treatment, a loss of doping by diffusion out of the surface during the drive-in diffusion has been neglected. This out-diffusion is low for boron, however significant for Al and Ga. Additionally, during drive-in diffusions often an

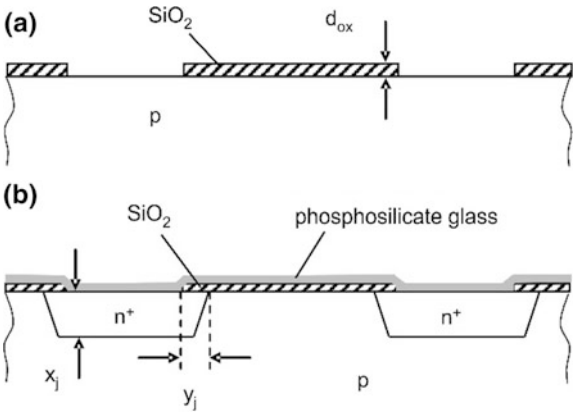


Fig. 4.20 Masking, example of fabrication of n^+-structures in a p-layer. **a** p-layer, SiO_2 structured with a photolithographic process. **b** Phosphorus diffusion with a penetration depth x_j

oxide layer grows at the surface and consumes some of the semiconductor. More detailed calculations of this process can be done with process simulation tools.

Masking with SiO_2 is not possible for the dopants Ga and Al, since for these impurities the diffusion constants in SiO_2 are too high.

Ion implantation is also often masked with SiO_2. The penetration depth of the ions into the oxide layer is in the same range as in Si because of the similar density. Therefore, the thickness of the oxide must be chosen accordingly. Other coating layers are also used for laterally masking during ion implantation, e.g. Si_3N_4 and even photoresist layers can be used, as long as the temperature during ion implantation is kept low.

4.7 Edge Terminations

The one-dimensional investigation of the blocking behavior in Chap. 3 is valid only as long as the semiconductor body is unlimited in lateral direction. In reality, however, the device structures have a finite dimension and an edge termination must be applied to lower the electric field at the edges. Edge termination structures may be divided in two main groups:

1. Edge structures with *beveled termination structures*. By beveling a defined angle can be adjusted between a lateral pn-junction and the surface, and thereby the edge is relieved from high electric fields. An overview is given in [Ger79].
2. Edge structures with a planar semiconductor surface are denoted as *planar termination structures*. An overview is given in [Fal94].

Beveled termination structures

The beveled edge contour is produced by mechanical grinding. The angle α is defined in relation to the junction from the high doped side to the low doped side. A beveled edge with negative bevel angle is shown in Fig. 4.21. The effect can be explained in a simplified way as follows. The equipotential lines of the space charge must intersect with the surface orthogonally, if there are no surface charges. This forces the space charge region to widen at the edge, and thereby the electric field strength is lowered at the surface.

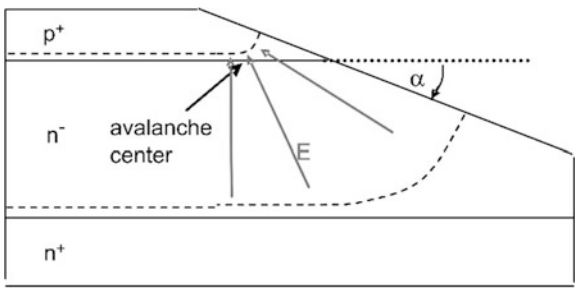


Fig. 4.21 Edge termination with negative bevel angle

In a structure with negative bevel angle, a compression of the field lines occurs close to the junction on the p-side near the beveled surface. Therefore, the electric field is increased at this location. To counteract this, an edge structure with negative bevel angle is always fabricated with a very shallow angle, usually between 2° und 4°. For this case, approximately 90% of the volume blocking capability can be achieved. The avalanche breakdown always starts at the edge close below the semiconductor surface at the location which is marked in Fig. 4.21 as avalanche center.

With a beveled edge termination using a positive bevel angle, as shown in Fig. 4.22, the distance between the equipotential lines is also increased at the surface. Especially close to the pn-junction, where the electric field is high, the field lines are widened at the edge. Therefore no avalanche center occurs and 100% of the volume breakdown voltage can be achieved with this termination structure. The angle α can be chosen in a wide range between 30° und 80°.

The etched structure shown in Fig. 4.23 is also a junction termination structure with positive bevel angle. The semiconductor wafer is etched starting from the n^+-side. Also for this structure, the breakdown will take place in the volume and the breakdown voltage is not lowered compared to the one-dimensional calculation.

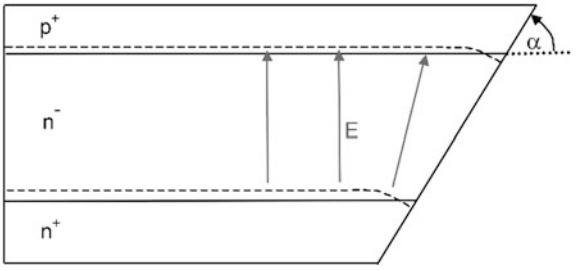


Fig. 4.22 Edge termination with positive bevel angle

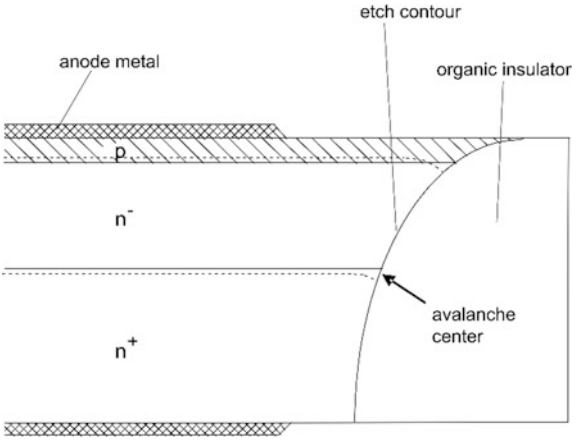


Fig. 4.23 Etched edge termination structure with positive bevel angle at the pn-junction

However, if the space charge penetrates the n^+-layer, an avalanche center may occur at the nn^+-junction at the location marked in Fig. 4.23.

If an avalanche center at the nn^+-junction is avoided, the structure according Fig. 4.23 proves to be very insensitive against surface charges. For the long time stability in this case a passivation layer of silicone gel is sufficient. The sharp corner at the anode side is mechanically very susceptible and such devices can easily be damaged. Therefore, this edge structure is not suited for modern devices with shallow penetration depth of the p-layer.

Planar junction termination structures

Planar structures are mechanically much more insensitive. The structure with floating potential rings, as shown in Fig. 4.24, can be fabricated in a single mask step together with the p-type anode layer. The potential rings lead to a widening of the space charge at the semiconductor top surface. The potential ring structure was first suggested by Kao and Wolley [Kao67].

The maxima of the electric field are marked in Fig. 4.24, they can be lowered by selecting the optimal distance between the potential rings. With numerical simulation using the Poisson equation in a two dimensional grid, the optimal arrangement of potential rings can be calculated, as shown by Brieger and Gerlach [Bri83]. Nevertheless field maxima cannot be completely avoided and the avalanche breakdown will occur in the region of the junction termination. Around 85–95% of the volume breakdown voltage can be achieved. A big advantage of this structure is that no additional photolithographic step is necessary in production; it accrues simultaneously with the fabrication of the p-layer, which is used as anode layer for a diode or p-base for a transistor. Therefore, this structure is the most frequently used edge termination structure. A disadvantage is the large space requirement.

With a very lowly doped p^--zone, the so-called Junction Termination Extension (JTE) structure, it is also possible to reach the volume breakdown voltage using planar structures. The Variation of Lateral Doping (VLD) structure shown in Fig. 4.25 is one of the possible varieties of the JTE structure, it was first suggested by Stengl und Gösele [Ste85]. A p^--zone, in which the doping is decreasing outwards to the device edge, is connected to the p anode layer.

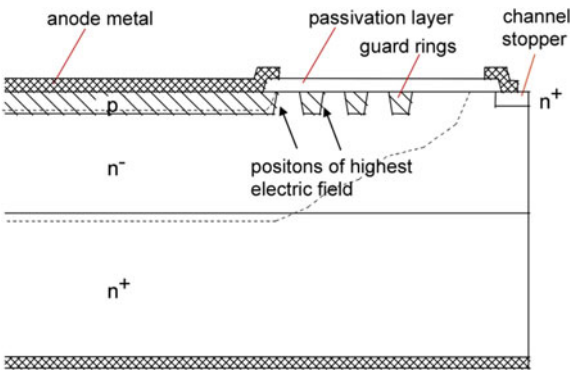


Fig. 4.24 Planar junction termination with floating potential rings

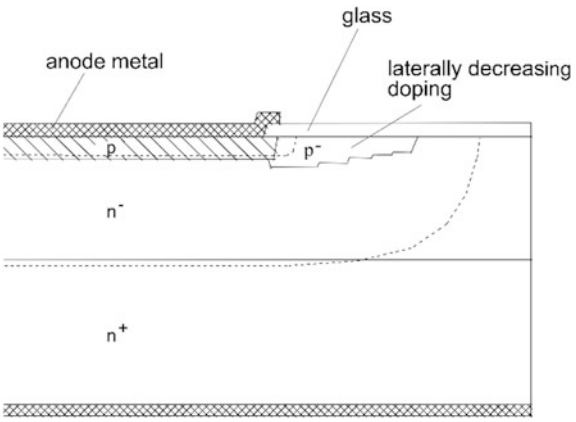


Fig. 4.25 Junction terminationedge termination structure with laterally decreasing doping (VLD)

The structure is manufactured in such a way that the mask for the p^--layer has stripe shaped openings, with a pitch in a form that the share of open area decreases from the anode region towards the edge. The deposition of the p-dopant is executed in these stripes, and during the following drive-in diffusion the p^--zones are joined by lateral diffusion. This process results in a profile with decreasing doping concentration and decreasing depth towards the edge as shown in Fig. 4.25. With an optimal design the breakdown takes place in the volume. Structures fabricated with implantation of Al reach 100% of the volume blocking voltage [Scu89]. Because of the lower solubility of Al, low-doped regions are easier to realize with Al.

Compared to the structure with floating potential rings, the VLD structure features a smaller space requirement and it is also much more insensitive against surface charge states [Scu89]. Because of the small tolerance window of the doping in the edge region, an ion implantation is necessary for controlling the deposited amount of B or Al. Regarding other parameters, e.g. the penetration depth, the VLD structure is less sensitive.

With field plate structures, the metallization of the p-layer is extended above the insulating passivation layer at the edge of the device. Figure 4.26a shows the effect of a one-step field plate. Also with this structure the space charge is elongated at the edge. A one-step field plate is rarely sufficient to reach blocking voltages close to the volume breakdown voltage. For the fabrication of field-controlled devices, several insulating layers are necessary in the cell structure, and using a combination of these layers several steps at the edge can be realized as shown in Fig. 4.26b. The determination of the position of the specific steps is done by numeric simulation solving the Poisson equation, similar to the calculation of field rings, mentioned earlier. Field plate structures are often used in MOSFETs and IGBTs.

It is also possible to combine guard rings and field plates in order to spare some potential rings and thus to reduce the area requirement.

For economic reasons, it is necessary to keep the edge termination area as small as possible, since this area cannot be used for current conduction. On the other hand, additional photolithographic steps should also be avoided. In several ways the

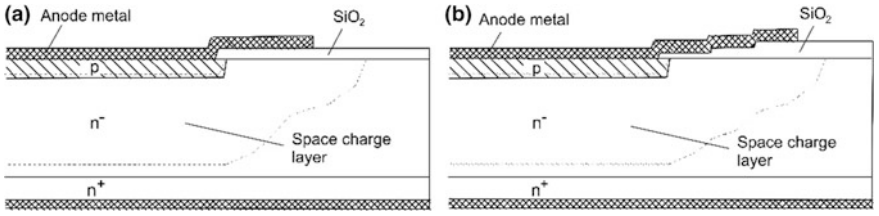


Fig. 4.26 Field plate junction termination. **a** One-step field plate. **b** Multi step field plate

edge is the weak point of a power semiconductor device and the development of protecting junction terminations is one of the most important tasks in the development of stable and rugged power devices.

Junction termination for bidirectional blocking devices

Bidirectional blocking devices like the thyristor have two blocking pn-junctions, where the low doped layer n^--layer is commonly used for both directions. Thyristors are fabricated for example with two structures of negative bevel angle like in Fig. 4.21, or with one structure of negative and one with a positive bevel angle as shown Fig. 4.22. Also other structures like etched mesa grooves from both sides are applied.

An attractive solution for a bidirectional blocking device is the structure with deep edge diffusion as shown in Fig. 4.27. The deep p-diffusions on the right side are carried out from both sides as one of the first process steps. In a final step the wafer is cut in the regions of the deep p-diffusions. The upper pn-junction has a planar junction with guard-rings and a channel stopper as in Fig. 4.24. The lower pn-junction is connected to the top side of the wafer via the deep p-layers. The deep p-diffusion diffuses additionally to the side and thus a p^--structure as used in JTE-structures is achieved. No further structures are necessary.

The advantage of the edge diffusion structure is that the wafer is compatible with processes of modern semiconductor technology, in which photolithographic processes can usually be applied only to one side of a wafer. Bidirectional blocking IGBTs are using a similar junction termination, see Sect. 10.7.

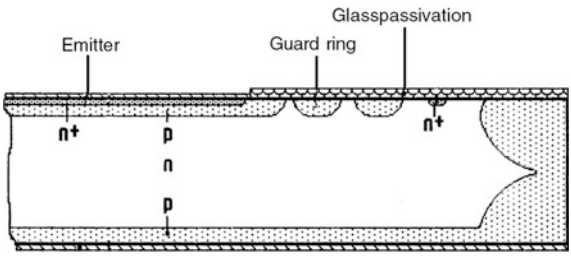


Fig. 4.27 Thyristor with edge diffusion. Figure from IXYS Semiconductors AG

4.8 Passivation

The surface of a semiconductor is very sensitive to high electric fields. In addition to the edge terminations just described, it is also necessary to treat the surface in some way to obtain a well-defined surface and to terminate the free bonds of the silicon atoms at the surface. This treatment is called surface passivation and involves cleaning processes and subsequent deposition of an insulating material or a material with high resistivity.

For conventional devices with beveled edge an organic passivation layer is often used which consists of silicon rubber or polyimide. For junction terminations like the beveled structure with positive angle (see Figs. 4.22 and 4.23) the passivation layer is not critical, since no field peaks occur at the interface.

In junction terminations such as the planar structure with potential rings, field peaks occur at the surface and the passivation becomes crucial. The resulting blocking capability of the device is very sensitive to charges in the passivation layer and the charge state of the passivating material must be taken into account in the calculation of the breakdown voltage.

Often SiO_2 is used as passivation layer. After diffusion processes, an oxidized semiconductor surface is attained often and no additional process for passivation is necessary. This oxide layer must have a very high purity and this requirement becomes more critical for lower background doping levels N_D, since at low N_D already a very small density of surface charges is sufficient to generate an inversion layer at the surface. Such inversion layers lead to increased leakage currents and long term instability of the device. Instead of SiO_2 also different glass layers are used, which consist of SiO_2 and additional elements.

Also semi-insulating layers are possible to use as a combined passivation and edge termination. By adjusting the electric conductivity of a semi-insulating layer, a continuous decrease of the potential at the surface can be achieved.

One commonly used evaluation method and selection criterion for the quality of a passivation layer is the hot reverse test (see Sect. 11.6), where devices are subjected to the maximal allowed temperature and the maximum allowed voltage, applied as DC voltage, for 1000 h. If there are mobile ions or other mobile charges in the passivation material, they will move, driven by the high electric field, and accumulate at unfavorable locations where an inversion layer may be created. In this case a significant leakage current increase will be observed and the device will be rejected.

Most challenging is the passivation of very high voltage devices between 5 and 10 kV, because of the very low doping which is mandatory for such devices. For these very demanding conditions a passivation layer of amorphous hydrated carbon (a-C:H) is used in some applications [Bar99]. The mechanical and chemical properties of a-C:H features a diamond-like characteristic, although the bandgap is smaller than for diamond, it is only in the range of 1–1.6 eV. One positive property of a-C:H is that mirror charges can be induced in the bandgap, which have the capability to compensate disturbing charges. They can even reduce field peaks at

the surface. Layers of a-C:H are very stable under the condition of hermitically sealed housings.

For SiC devices, ten times higher fields are allowed in the bulk, as compared to silicon. This imposes difficult challenges for SiC passivation layers and new solutions are required both for surface processing and for the insulating layers.

4.9 Recombination Centers

Fundamental device characteristics depend on the charge carrier lifetime. The carrier lifetime in silicon, as well as all other indirect bandgap semiconductors, is adjusted by recombination centers. Recombination centers always decrease the carrier lifetime and therefore increase the voltage drop at forward conduction of a device, which leads to higher on-state losses. On the other hand, recombination centers decrease many parameters important for the switching behavior, for instance the switching time and the reverse recovery charge. This lowers the losses associated with the turn-off. The following part of this chapter deals with the particularities of different recombination centers, for instance their temperature dependence and how they can affect device characteristics. For power devices it is very important to know the differences between different recombination center technologies and how they can be used to optimize the trade-off between on-state and switching losses.

Gold and platinum as recombination centers

Gold is the earliest used recombination center in silicon [Far65]. Platinum is used as a recombination center since middle of 1970s as an alternative to gold [Mil76]. Both recombination centers, their energy levels (Fig. 2.19) and physical characteristics have been discussed in Sect. 2.7.

Gold as well as platinum are diffused into silicon, and both have similar characteristics in their diffusion mechanism. In silicon they can occupy interstitial and substitutional positions, however, the solubility of the heavy metal is higher at the substitutional sites. On the other hand, gold and platinum on interstitial sites are much more mobile and diffuse rapidly. The diffusion from substitutional heavy metals can in fact be neglected, but there is always a relatively small barrier for interstitial to substitutional sites which results in a very fast diffusion. For instance, at a diffusion temperature of 850 °C, a substantial part of the heavy metals can be found at the opposite side of the semiconductor wafer already after a diffusion time of only 10 min. This fast diffusion leads to an U-shaped concentration versus depth, or a bathtub-shaped profile with increased density of heavy metal atoms close towards the wafer surfaces, as shown in Fig. 4.28. This profile also means that the density is increased close to the pn- and nn^+-junctions in devices.

Figure 4.28 does not contain highly doped zones, which are produced in power devices before the gold diffusion. Highly boron-doped p-layers will have more

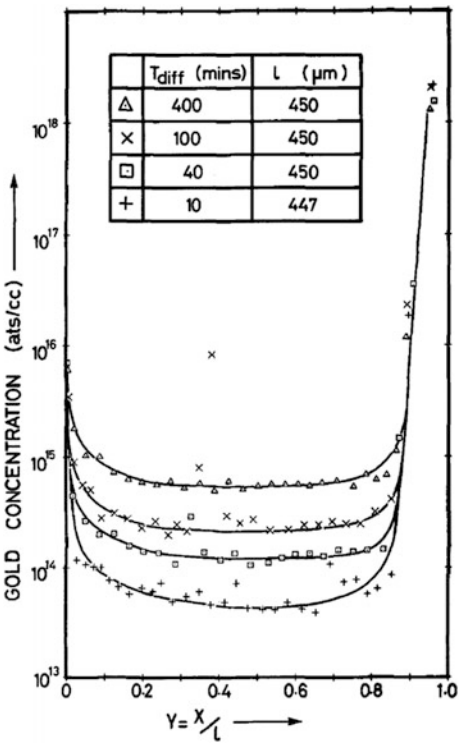


Fig. 4.28 Concentration profiles of gold diffused into FZ silicon slices at 900 °C. Gold deposition before diffusion was executed on the right-hand side. The very high concentrations on this side are not valid. Figure from [Hun73] © 1973 The Electrochemical Society

stress in the lattice and will exhibit an increased gold concentration. Highly phosphorous-doped n^+-layers will collect gold atoms (gettering), a gold diffusion through such layers is not possible.

It is difficult to control the diffusion profile of gold and platinum. Because of the interaction of the diffusion mechanism with other defects in the crystal, it is also difficult to obtain a good reproducibility of the diffused heavy metal profiles. Over many years of application of gold and platinum diffusion to control the charge carrier lifetimes, a wide scattering in the device characteristics even within the same batch, as well as a bad yield had to be accepted. Much due to the problems to control the gold and platinum diffusion, a large difference is fond between the real parameters and the maximal allowed values in data sheets of many fast diodes of older generations.

While the diffusion mechanism is very similar for gold and platinum, the characteristics of gold and platinum diffused devices are very different.

Platinum features a decrease of the lifetime with the injection level, see Fig. 2.21, this holds not for gold. Using gold, it is possible to achieve a relative good trade-off between forward voltage drop and reverse recovery charge extracted at turn-off. In this respect, gold would be a suitable recombination center. However, one of the Au levels is located almost exactly in the middle of the bandgap, see Fig. 2.19. This has the consequence that it acts also very effectively as generation center during the

blocking state and this generation creates a high reverse leakage current at elevated temperature, for further details see Fig. 3.13. The leakage current of devices, where gold has been diffused to control the recombination rate, are at 150 °C a factor of 50 higher than for platinum diffused devices.

The leakage current for different recombination center technologies are compared in Fig. 4.29. For devices with a high density of gold recombination centers as necessary for instance for fast freewheeling diodes and for a rated voltage over 1000 V, at a temperature of 150 °C such high reverse blocking losses occur that they will, in fact, result in thermal instability. Gold must therefore be excluded as recombination center for fast freewheeling diodes in this voltage range. As a thermal limit in Fig. 4.24, it is assumed that the temperature increase by the leakage current is allowed to be at maximum $\Delta T = 15$ K at a DC voltage of 2/3 of the specified blocking voltage, and a thermal resistance $R_{thjc} \times A = 31\ \mathrm{Kmm^2W^{-1}}$ is assumed which is, for example, a typical value for a power module.

For gold-diffused devices at a voltage above 1000 V, the maximum allowed junction temperature is limited, typically to 125 °C. For a DC voltage load it must be further reduced and limited to only 100 °C.

In a neutral n-region the gold atoms are mostly negatively charged because the Fermi level lies above the acceptor level of gold. If the density of the gold atoms is in the range of the background doping – this is the case for very fast power diodes – then a compensation effect occurs and the device behaves according to the reduced background doping [Mil76, Nov89]. This affects also the turn-on behavior. The voltage peak V_{FRM}, occurring at turn-on of a device, is a function of the resistance of the low-doped layer – see Chap. 5, Eq. (5.66). For gold diffused devices, V_{FRM} can amount to the multiple of the voltage peak of diodes completely without, or with other recombination centers.

For platinum, the trade-off between forward voltage drop and reverse recovery charge is significantly less convenient than that for gold. On the other hand, for platinum there is no energy level close to the middle of the bandgap, and the small

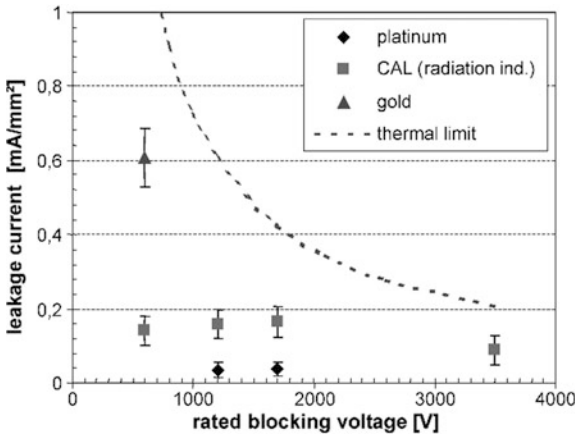


Fig. 4.29 Leakage current of diodes fabricated with different recombination centers at $T = 150$ °C. *X*-axis: Rated voltage of the respective device. The density of recombination centers is chosen so that requirements for freewheeling diodes for IGBTs with the specified voltage are fulfilled. Figure from [Lut97] © 2007 EPE

leakage current of a platinum diffused device can hardly be distinguished from a device without recombination centers. Therefore higher junction temperatures can be realized with platinum, for instance 150 or 175 °C and in the future probably 200 °C.

For platinum diffused fast diodes the reverse recovery charge is strongly increasing with temperature. The effect of the recombination center is therefore decreasing with increasing temperature. As a consequence platinum diffused diodes, in which the p-emitter is strongly doped and the emitter recombination is low, show a negative temperature dependency of the forward voltage. The temperature dependency of capture coefficients of different recombination centers is summarized in [Sie01].

Radiation induced recombination centers

The characteristics of the most important centers generated by radiation have been discussed in Sect. 2.7. This technology has a much better reproducibility, and it is easy to control the lifetime by adjusting the radiation dose. The process of irradiating silicon power devices with high energy electrons to induce lattice defects, which act as recombination centers, has been in use since the 1970s. Electron and also and γ-radiation creates a homogeneous distribution of defects by knocking out lattice atoms in random collisions with low probability. This technique has many advantages over diffusion technology, but the resulting constant distribution of recombination centers is often a disadvantage for the switching characteristics of devices. More recently, implantation of protons or helium ions has been introduced for charge carrier lifetime control [Lut97, Won87]. These particles also collide with the host atoms and knock them out of their positions, but the collisions occur with large probability towards the end of the ion track. As a result, low mass ion implantation creates a localized region of recombination centers, whose position can be adjusted by the energy of the incoming particles. This process has become widely used for the adjustment of charge carrier lifetime (see also Sect. 5.7, CAL diode). Irradiation processes feature high accuracy and high reproducibility and, furthermore, the irradiation techniques offer the possibility to perform the carrier lifetime control at the end of the fabrication process after the devices have been metalized and passivated.

The vacancies and interstitials that are generated by the collisions diffuse and form stable complexes with each other and with impurity atoms which are present even in high purity silicon, such as carbon, oxygen and phosphorus. The irradiation process is followed by a mild annealing step to eliminate the thermally unstable centers and to ensure a long-term stability of the device characteristics. The annealing process is also needed to reduce the effects of further processing steps, such as soldering etc., that can alter the defect properties.

Devices for packaging in press pack technology are annealed in the range of 220 °C and they will not be exposed to a higher process temperature after the irradiation. Devices which are exposed to a soldering process during packaging and other possible thermal treatments should be annealed at temperature in the range

340–350 °C to ensure that no further defect annealing occurs in subsequent soldering processes, whose temperatures can be quite high.

A lot of work has been done to determine the characteristics of the radiation induced centers [Ble96, Hal96, Scu02, Sue94], a summarizing work of recent results is given in [Sie06] and [Haz07]. The most important centers are shown in Fig. 2.22. Figure 4.30 shows additionally the centers which are still present after annealing at 220 °C and which impact the carrier lifetime or other device characteristics [Sie02, Sie06]. The notations in the base line E(90 K) denote the signal of the center found in Deep Level Transient Spectroscopy (DLTS). This notation is often used, since the determination of the specific atomic structure is difficult and the literature on it is not always consistent.

The OV-center or A-center E(90K) is a vacancy-oxygen complex. This center starts to anneal at temperatures above 350 °C and vanishes almost completely at an annealing temperature of 400 °C [Won85]. As described in Sect. 2.7, the high-injection lifetime τ_{HL} and the resulting forward characteristics as well as switching properties are mainly controlled by the OV-center.

The K-center H(195K) was found as a hole trap at +0.35 eV above the valance band. Different assignments on its origin are fond in the literature, often a COVV-complex involving a carbon impurity, an oxygen atom and two vacancies is assumed. Recent work using the Cathode Luminescence method identified the relevant center as C_iO_i, an association of an interstitial carbon atom and an interstitial oxygen atom [Niw08]. The K-center starts to anneal at a temperature in the range of 370–400 °C and vanishes after annealing at temperatures over 450 °C [Won85]. It has an energy level close to the middle of the bandgap, but the contribution to recombination processes is rather small due to low probability to capture charge carriers, i.e. small capture cross sections. Under high-injection condition, which is given in a forward-biased power diode for example, this center is positively charged. When the diode is turned-off sufficiently fast, the defects remain positively charged for a certain time, typically some 100 ns or 1 µs, due to the relatively low electron-capture rate. The positively charged K-centers will act as donors, and they will increase the effective doping concentration. This reduces temporarily the breakdown voltage at the pn-junction according to Eq. (3.84), resulting in avalanche breakdown at voltages far below the static breakdown

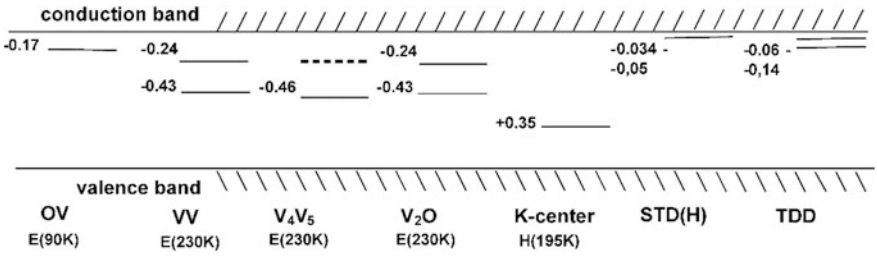


Fig. 4.30 Energy positions in the bandgap for the most prominent irradiation-induced defects in silicon

voltage [Lut98]. These detrimental effects are further described in Sect. 13.3. It is thus important to avoid high densities of K-centers, and the maximum irradiation dose is therefore limited for some applications.

The divacancy, VV E(230K), has as many as three different charge states, that give rise to three separate levels in the bandgap. Most important for charge carrier dynamics is the level at 0.43 eV below the conduction band edge. It acts as a recombination center, but since this energy level is so close to the middle of the bandgap, it also acts as a generation center and will determine the generation lifetime and leakage current. However, its effect as generation center is fortunately weaker than gold.

Divacancies anneal out at temperatures of about 300 °C [Bro82, Won85], but when using proton- or $^4He^{2+}$-irradiation, this level is still found even after annealing temperatures in excess of 350 °C. This can be explained by a transformation of the defect to another defect with nearly the same bandgap position, and the remaining centers have been assigned to the singly- and doubly-charged states of the V_2O defect [Mon02]. Another possible origin of this E(230K) peak is given in [Gul77], which suggests that it arises from a V_4 or V_5 complex. Because of their low concentration, the influence of the V_2O defects on τ_{HL} is rather small, but this center is responsible for the fact that devices implanted with helium ions exhibit a higher leakage current than platinum diffused devices (see Fig. 4.29). The leakage current of helium-implanted devices is about 20% of the leakage current of comparable gold diffused devices and poses no problem for the thermal stability at junction temperatures up to 150 °C.

The E(230K) defect also shows a significant compensation effect on the doping in n-type silicon, where the Fermi-level is above the trap level W_C −0.43 or −0.46 eV. After annealing at a temperature of 350 °C, one can find a decrease of the effective doping due to the charged acceptor-states of E(230K) in the region affected by the helium-implantation. This effect can be used to increase or to adjust the blocking voltage of a device after the main manufacturing steps [Sie06].

Annealing temperatures significantly above 350 °C may result in the formation of so called thermal double-donors TDD. The maximum TDD concentration was found after annealing at $T \approx 450$ °C. The TDDs increase the doping concentration in n-type silicon but compensate the doping in p-type silicon.

While radiation induced centers are used widely, their atomic structure and the charge carrier capture and emission processes are still not fully understood. An overview with a detailed treatment of the characteristics of the specific centers can be found in [Sie06]. Many important details, for instance the temperature dependency of the center characteristics, are still an object of active research.

Radiation enhanced diffusion of Pt and Pd

As described, the diffusion mechanism as well as the built-in mechanism of Au and Pt into the Si lattice interacts with crystal defects. Therefore the final profile can be influenced if there is defined crystal damage at a defined penetration depth as it is created by radiation with particles such as H^+ and He^{++}-ions. After a He^{++}-

irradiation, Pt can be diffused at a temperature much lower than usual for Pt-diffusions into the position of maximal radiation damage caused by the He^{++} implantation [Vob02]. If the temperature is high enough to anneal the radiation-induced defects, they are replaced by a local profile of Pt atoms. Since not all of the radiation induced defects have optimal properties, the final device characteristics can be further improved by the radiation enhanced diffusion. A level close to the middle of the bandgap, which leads to increased leakage current, is avoided if Pt or Pd is used.

The energy levels of Pd are at positions in the bandgap close to that of Pt, so that the electronic structure of a substitutional Pd atom seems to be quite similar to a substitutional Pt atom. A wider usable temperature range was found for the radiation enhanced diffusion of Pd [Vob07]. Depending on the temperature, the formation of an acceptor is observed and a buried p^--layer can be formed at the penetration depth of the primary He-implantation [Vob09]. This p^--layer increases the static breakdown voltage and reduces peaks of the electric fields and dynamic avalanche at fast switching events of diodes. Diodes with increased ruggedness have been presented using this technique [Vob09]. Further work is of interest to understand the details of the created centers.

4.10 Radiation-Induced Doping

After proton irradiation, additional effects arise since the protons (hydrogen ions) may also participate in the defect complexes [Gor74, Ohm72]. The hydrogen-related shallow thermal donors STD(H) are an example for this. These centers are found after annealing of a proton implanted sample above 200 °C [Won85]. Their maximum density is found in the region of the projected range R_{pr} of the protons, a clear indication that the defects involve the implanted hydrogen. The STD(H) is a stable donor with an energy level close to the conduction band. If proton implantation is used to reduce the carrier lifetime, the dose must be limited to keep the density of these hydrogen related centers below the background doping. Otherwise, deep buried n-layers are created. However, this is also used for intentionally doping to adjust the threshold voltage of protection devices [Sie06].

The relationship between proton dose and created donors is shown in Fig. 4.31 taken from [Klu11].

For FZ-silicon, the following equation expresses not too high doses by [Klu11]

$$N_D[\mathrm{cm}^{-3}] = (12.6 \pm 0.8)\mathrm{cm}^{-1} \times \Phi\ [\mathrm{cm}^{-2}] \tag{4.24}$$

At about 10^{16} cm^{-3}, the amount of created donors reaches an upper limit. In [Klu11] it is shown that not only hydrogen but also oxygen is involved in the created center and the maximum density is limited by the oxygen content.

If proton implantation is executed in conjunction with an adjacent subsequent annealing step at temperatures between 300 and 500 °C, “buffers” which lead to a

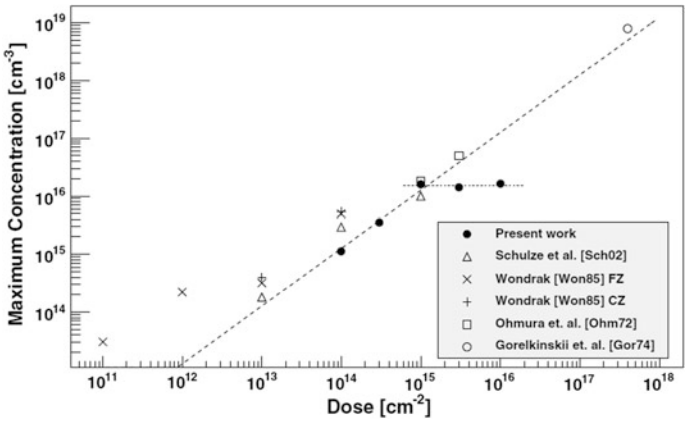
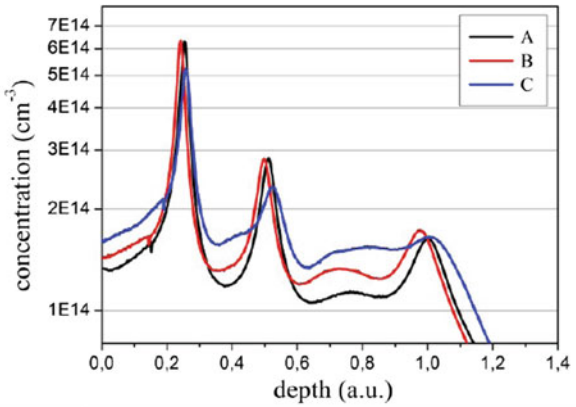


Fig. 4.31 Maximum peak charge carrier concentration for 2 meV protons. The dashed line shows the assumed linear relationship. The upper limit found for high doses is indicated by the dotted line. Fig. from [Klu11]

trapezoidal field in the device (PT-dimensioning, Chap. 5) can be created. This technology is applied for some IGBTs of the latest generations.

In [Scu16] it is shown that, at presence of H, the radiation-induced C_iO_i centers transform into donor-like electrically active C_iO_i-H complexes at annealing above 350 °C. The resulting profile depends on the carbon content, see Fig. 4.32.



Impurity (cm^{-3})	Substrate		
	A	*B*	*C*
oxygen	1.9×10^{17}	2.4×10^{17}	2.1×10^{17}
carbon	5.4×10^{14}	7.5×10^{14}	2.4×10^{15}

Fig. 4.32 Profile of radiation-induced donors for a three-step proton implantation in magnetic CZ silicon, depending on carbon and oxygen content of the substrate. © 2016 IEEE. Reprinted, with permission, from [Scu16]

The profile in Fig. 4.32 was achieved with a three-step irradiation of magnetic CZ-Silicon. The peaks at the maximum penetration of hydrogen are visible, and a slowly increased doping towards the surface, as intended, is created. Immediately after irradiation, hydrogen-associated donors are created only in a narrow region around the projected range R_{pr} in accordance with the distribution of the hydrogen atoms. Lattice defects, however, are present up to the surface. Due to the charged acceptor-states of E(230K) in the region affected by lattice defects, a compensated or even nearly p-type region can occur. During subsequent annealing, a diffusion of hydrogen towards the surface takes place. The intensity of intended donors depends on the carbon content, see Fig. 4.31. Therefore, all process steps have to be developed carefully respecting the initial carbon and oxygen content of the wafer.

Radiation-induced donor doping needs, compared to diffusion processes, only a low temperature at annealing. Therefore, this method is applicable for thin-wafer technology even for a wafer diameter of 30 cm. Based on magnetic Czochralski-wafers, this leads to a very effective production process of IGBTs and freewheeling diodes.

4.11 Some Aspects on Technology of GaN Devices

As reported in the introduction, the semiconductor GaN is with respect to bandgap with 3.4 eV and the critical field for avalanche similar to 4H-SiC, also the mobility of electrons is similar. It can be doped to n- and p-type conductivity, but it is mostly not intentionally doped (see below). Contrary to SiC, large area single crystals of GaN are not available, although significant progress in the growth of single crystal GaN has been made in recent years. GaN power devices as far as developed till now are based on a high quality crystal layer of GaN grown on different substrates. The devices consist actually of a hetero structure with AlGaN, which is an essential constituent deposited on the GaN as a thin film. Because the technology of GaN power devices differs widely from devices in Si and SiC, some aspects of the technology are described in the present section.

GaN crystallizes apart from a metastable cubic zinkblende polytype in the stable hexagonal wurtzite lattice, which is used for devices. The substrates must match with this lattice to some degree, to enable the hetero-epitaxy without creating a huge density of crystal defects not admitted for high-performance devices. Besides the lattice, the coefficient of thermal expansion (CTE) should not much differ from that of GaN in order to limit mechanical stresses caused by cooling down from the temperature of epitaxy. In Fig. 4.33 the CTE and the relevant lattice parameter are plotted for GaN itself and the substrate materials AlN, 4H-SiC, silicon and sapphire (monocrystalline Al_2O_3).

Like GaN, AlN crystallizes in the wurtzite lattice, and also the other substrates have a hexagonal symmetry, except silicon, for which this holds however within the (111) plane, the surface used for the epitaxial growth. As is seen, AlN and 4H-SiC come nearer to GaN than Si and sapphire. Nevertheless, from cost reasons Si is the mainly used substrate for power devices, while optoelectronic devices, e.g. light

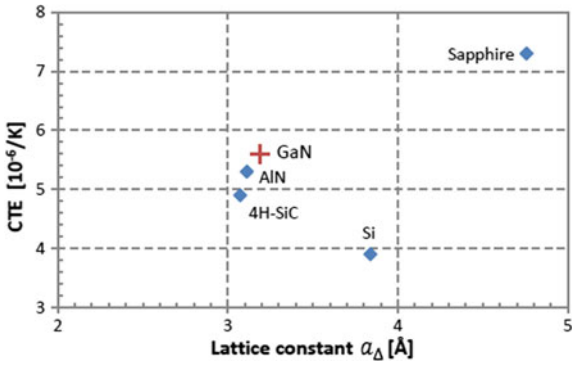


Fig. 4.33 Coefficient of thermal expansion and lattice constant a_Δ for GaN and several substrate materials. a_Δ is the distance between nearest lattice points within the c-plane or within the (111) plane for Si. In the latter case $a_\Delta = a_\square/\sqrt{2}$ where $a_\square$ is the side length of the cubic unit cell. The CTEs are mean values over the range of 20–800 °C and for the hexagonal crystals hold for the c-plane [Rod05, Yim74, LiB86, Swe83]

emitting diodes (LEDs), are primarily produced using sapphire as substrate. Some microwave devices are produced with GaN on SiC because of the better thermal conductivity of SiC. AlN is used as a thin interface layer (see below).

The use of Si as substrate implicates that cooling down after epitaxy generates high thermal strain. The GaN layer tends to constrict itself stronger than the substrate, and beyond a critical point it cracks. To reduce the strain the temperature of epitaxy is chosen as low as possible, typically in the range 740 to 800 °C. It turned out that high quality GaN layers could be grown on silicon only by using an intermediate buffer layer of $Al_xGa_{1-x}N$, in which Ga ions are partly replaced by Al [Krü98]. The layers are deposited from a gaseous composition of Triethylgallium and Trimethylalluminium and pure GaN or AlN (Metal-Organic Chemical Vapor Deposition method, MOCVD). During the deposition of the buffer layer the wafer gets a convex bowing, which is chosen in a manner that it is just compensated by the strain produced during cooling. By these measures of 'stress engineering' the final wafer bow must be reduced below about 50 µm in height, this is required for the following manufacturing steps, especially the photolithographic alignment [Ger15].

A typical graded structure of a GaN-on-Si wafer is shown in Fig. 4.34. On Si, a thin layer of AlN is deposited to improve the nucleation for the following low-temperature epitaxy. Then the buffer layer of $Al_xGa_{1-x}N$ is deposited, where x decreases from 0.75 to 0.25. The bandgap of this material varies between the bandgap of AlN with 6.2 eV and that of GaN as determined by the parameter x. Besides the purpose of strain compensation this layer has the function to isolate the drain of the device from the substrate, which is preferably on source potential. Hence the buffer layer is intentionally not doped. If necessary, the resistivity is enhanced by compensation with deep impurities. Also its thickness has to comply with the requirement of blocking. For a 600 V switching transistor, a buffer

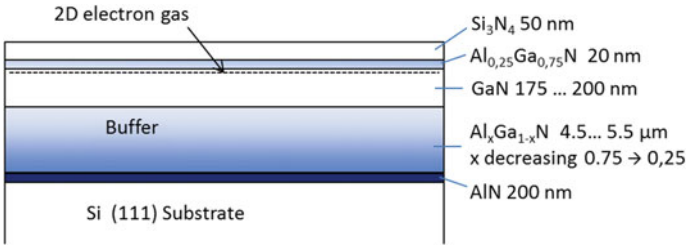


Fig. 4.34 GaN-on-Si wafer as used for lateral high-electron mobility transistors and Schottky-diodes. Drawing according to information from EpiGaN

thickness of minimum 4.5 µm is required. Often a further buffer layer not shown here is used consisting of several thin layers of GaN and AlN, forming a so-called super-lattice buffer [Ued05, Ued17].

The GaN layer following the buffer layer is the active channel layer used for switching. Its thickness is for example 175–200 nm. Thickness and n-doping must be low enough that it can be depleted with a gate voltage. An essential part of the channel is a highly conducting electron layer, termed as two-dimensional electron gas (2DEG), which is induced by the high electric polarization of GaN, as discussed below. The next following layer consisting of $Al_xGa_{1-x}N$ with typically $x = 0.25$ is called barrier layer. Its thickness is in the 20 nm range.

GaN and to a still larger extent AlGaN exhibit a high electric polarization. This is visualized by Fig. 4.35, which shows the wurtzite lattice structure of GaN. The crystal consists of stacked bilayers, one plane of which is closely packed with Ga and the other with N atoms. Nitrogen is an element with high electronegativity: Instead of spending an electron for a covalent bond it has a strong tendency to attract electrons from Ga to complete its outer electron shell. Hence the Ga-N bonds are to an essential degree of ionic or polar character. The result is that the surface of the crystal consisting of Ga atoms (Ga-face, upper surface in Fig. 4.35) has a positive charge and the other surface consisting of N atoms (N-face) a negative, so

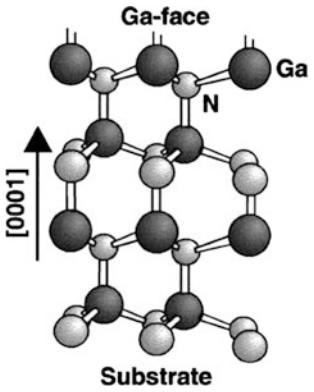


Fig. 4.35 Schematic drawing of the crystal structure of wurtzite GaN, Ga-face. Reproduced from [Amb99], with the permission of AIP Publishing

the crystal is strongly polarized. The polarization is equal to the sheet surface charges and directed by definition from the negative to the positive charge. The polarization of AlN is even nearly 3 times stronger than that of GaN, and this transfers to $Al_xGa_{1-x}N$ in proportion to the share x of Al. In addition to the 'spontaneous' polarization without strain there is a considerable piezoelectric polarization part caused by the strain of the layers resulting from the mismatch of the lattice constants and the coefficient of thermal expansion.

A precondition for the creation of the 2DEG in the GaN is that the needed carriers are generated sufficient quickly in the volume or flow to from the contacts at the sides. In contrast to GaN the barrier and buffer layer of AlGaN can be assumed essentially as isolators without available carriers. If now the GaN layer is grown on the buffer with the Ga-face at the top, also the AlGaN barrier layer will grow on it with the same orientation, i.e. with the N-face at the interface and the AlGa-face at the top. The fixed charge at the interface is made up by the positive charge of the Ga-face of the GaN layer and the higher negative charge density of the N-face of the AlGaN barrier film, the net interface charge is hence negative. In addition to the interface charge also the positive charge of the AlGa-face at the top of the barrier layer has a strong influence. This is illustrated in Fig. 4.36, which shows the surface and interface charges and the resulting field distribution in the barrier and the adjacent part of the channel layer. The charge of the AlGa-face neutralizes the charge of the N-face of the AlGaN-layer. Hence only the positive charge of the Ga-face of GaN remains in effect to attract electrons in the GaN layer. The jump of the field at the surfaces is q/ε times the sheet charge density there (The dielectric constant has practically the same value $\varepsilon = 8.9$ in both layers). The field in the AlGaN film is very high. The field in the GaN is a measure for the number of electrons existing behind a given point, see the next passage. The main result is that the charge of the Ga-face of the GaN layer, which is equal to the net charge at the surface of AlGaN and the interface, determines the sheet charge density of the

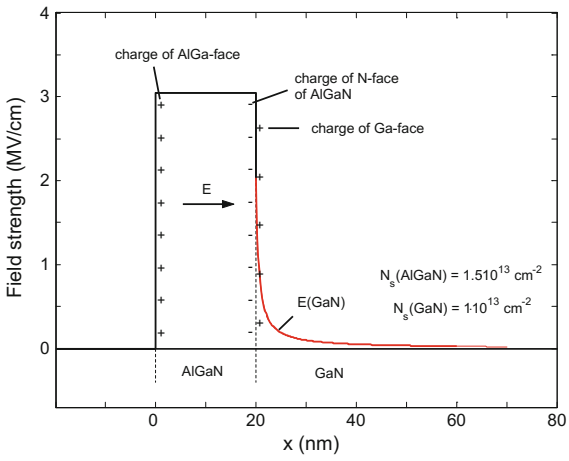


Fig. 4.36 Surface and interface charges and field distribution in the AlGaN barrier layer and the adjacent part of the GaN channel layer. The surface charge densities of AlGaN are assumed to be $N_s = \pm 1.5 \times 10^{13}$, that of the GaN layer 1×10^{13} cm^{-2} (see text)

induced 2DEG. This will hold also, if the field in the AlGaN-layer is not maintained because carriers are still present.

The 2DEG is known from MOSFETs as accumulation layer. Its extension in the depth can be calculated analytically, if the doping concentration is negligible. By double integration of the Poisson equation using Boltzmann statistics one obtains the following dependency (x coordinate perpendicular to the interface):

$$n = \frac{n(0)}{(1 + x/L_D)^2}, \tag{4.25}$$

where L_D is the Debye length at half of the surface concentration $n(0)$:

$$L_D = \sqrt{\frac{2\,\mathrm{kT}\varepsilon}{q^2 n(0)}}. \tag{4.26}$$

At $n(0) = 1 \times 10^{20}$ cm^{-3} Eq. (4.26) yields for GaN with $\varepsilon = 8.9 \cdot \varepsilon_0$: $L_D = 0.504$ nm. The sheet concentration is obtained from (4.25) as $N_s = n(0) \times L_D = 5.04 \times 10^{12}$ cm^{-2}. Although 2/3 of the electrons are contained in the range $x \leq 2L_D = 1.01$ nm, the concentration at $x = 30$ nm is still 2.7×10^{16} cm^{-3}. The extension is comparable to the thickness of the inversion channel of a silicon MOSFET. The example agrees nearly with the case of Fig. 4.36. The field in GaN was calculated there using (4.25) from

$$\mathsf{E} = -\frac{\mathrm{kT}}{q}\frac{d\,\ln n}{dx} = \frac{E(0)}{1 + x/L_D}.$$

The polarization of GaN amounts up to 2×10^{13} elementary charges per cm^2, an equal number of mobile electrons is induced in GaN. Compared with a silicon MOS-device this is very high: For an oxide thickness $d_{ox} = 50$ nm, a threshold voltage $V_T = 5$ V and an applied gate voltage $V_G = 15$ V, one obtains from Eqs. (9.1) and (9.2) later in Chap. 9.4 a sheet electron density of 4.3×10^{12} cm^{-2} in the channel.

The mobility of the electrons in the 2DEG is found to be higher than the channel mobility in silicon MOSFETs. Values ranging from 1000 to 2000 cm^2/(Vs) are reported at room temperature for a sheet carrier density of 1×10^{13} cm^{-2} [Amb99]. The spread is explained by a varying surface roughness. The mobility $\mu_{n(\mathrm{Ch})}$ in Si IGBTs or MOSFETs is under typical conditions in the range 300–500 cm^{2}V^{-1}s^{-1}. Electron mobilities in bulk GaN are found to be up to 990 cm^{2}V^{-1}s^{-1} [Qua08]. The carriers in the 2DEG (as well as in a MOSFET channel) are spatially separated from dopants, hence scattering at impurities is less important than in the neutral bulk. For SiC MOSFETs the best values reported are about 70 cm^{2}V^{-1}s^{-1}, compare Chap. 9. Due to both the high sheet concentration and the high electron mobility, the 2DEG in GaN is very effective in current conduction.

If necessary, n-type doping of GaN can be established by silicon as dopant. The activation energy ΔE_G of silicon in GaN is 5–9 meV allowing effective ionization [Qua08]. For p-type doping Mg is mostly used, the ionization energy is reported to be 173 meV [Qua08]. Unintentionally doped GaN is typically of n-type with electron concentration $n = N_D - N_A$ of the order 10^{16} cm^{-3} [Tan10]. The n-type conductivity is attributed to oxygen incorporation [Van04]. Also nitrogen vacancies are shallow donors, whereas Ga vacancies are triple charged acceptors [Van04]. The little controllable doping of GaN is still a problem.

Meanwhile, 8-inch GaN-on-Si wafers can be produced. The GaN-on-Si wafers can be used for device fabrication in a silicon CMOS production line in CMOS foundries, using the standard tool set for creating structures. To avoid cleanroom contamination by free-floating GaN particles, a Si_3N_4 passivation layer which is about 50–100 nm thick is of advantage. It promises a smooth and contamination-free surface [Beh15]. Because of the crystalline mismatch and different CTE, GaN-on-Si wafers are more fragile compared to silicon wafers. A GaN-on-Si wafer uses a thicker substrate as it is the case for Si in CMOS-lines [Beh15]. The functioning of GaNdevices is discussed in Sect. 9.12.

References

[Amb99] Ambacher, O., et al.: Two-dimensional electron gases induced by spontaneous and piezoelectric polarization charges in N- and Ga-face AlGaN/GaN heterostructures. J. Appl. Phys. **85**(6), 3222–3233 (1999)

[Amm92] von Ammon, W.: Neutron transmutation doped silicon – technological and economic aspects. Nucl. Instrum. Methods Phys. Res. **B63**, 95–100 (1992)

[Bar99] Barthelmeß, R., Beuermann, M., Winter, N.: New diodes with pressure contact for hard-switched high power converters. Proceedings of the EPE '99, Lausanne (1999)

[Beh15] Behet, M.: GaN—Promise to Reality: The Next Generation of Power Electronics is Taking Shape. Display + web Publication, http://www.displayplus.net/news/articleView.html?idxno=64606. Download 6 Jan 2017

[Ben99] Benda, V., Govar, J., Grant, D.A.: Power Semiconductor Devices. Wiley, New York (1999)

[Ble96] Bleichner, H., Jonsson, P., Keskitalo, N., Nordlander, E.: Temperature and injection dependence of the Shockley–Read–Hall lifetime in electron irradiated n-type silicon. J. Appl. Phys. **79**, 9142 (1996)

[Boe11] Böving, H., Laska, T., Pugatschow, A., Jakobi, W.: Ultrathin 400V FS IGBT for HEV applications. In: Proceedings International Symposium on Power Semiconductor Devices and ICs ISPSD 2011, pp. 64–67 (2011)

[Bor87] Borisenko, V.E., Yudin, S.G.: Steady-state solubility of substitutional impurities in silicon. Phys. Status Solidi (a) **101**, 123–127 (1987)

[Bri83] Brieger, K.P., Gerlach, W., Pelka, J.: Blocking capability of planar devices with field limiting rings. Sol. State Electron. **26**, 739 (1983)

[Bro82] Brotherton, S.D., Bradley, P.: Defect production and lifetime control in electron and γ-irradiated silicon. J. Appl. Phys. **53**(8), 5720–5732 (1982)

[Bul66] Bullis, W.M.: Properties of gold in silicon. Solid-State Electron. **9**, 143–168 (1966)

[Cho11] Chowdhury, I., Chandrasekhar, M., Klein, P.B., Caldwell, J.D., Tangali, S.: High growth rate 4H-SiC epitaxial growth using dichlorosilane in a hot-wall CVD reactor. J. Cryst. Growth **316**(1), 60–66 (2011)

[Cla11] Clark, P.: IMEC plans 450 mm wafer fab module for 2015. EETimes.com, October 11 (2011)

[Cra56] Crank, J.: The Mathematics of Diffusion, p. 148. Clarendon Press, Oxford (1956)

[Dea68] Dearnaley, G, Freeman, J.H., Gard, G.A., Wilkins, M.A.: Implantation Profiles of ^{32}P Channelled into Silicon Crystals. Can. J. Phys. **46**, 587ff (1968)

[ECP16] Friedrichs, P. et al.: ECPE position paper on next generation power electronics based on wide bandgap devices—challenges and opportunities for Europe http://www.ecpe.org/roadmaps-strategy-papers/strategy-papers/ (2016)

[Fai81] Fair, R.B.: Concentration profiles of diffused dopants. In: Wang, F.F.Y., (ed) Impurity doping processes in silicon, North Holland (1981)

[Fal94] Falck, E.: Untersuchung der Sperrfähigkeit von Halbleiter-Bauelementen mittels numerischer Simulation, Dissertation, Berlin (1994)

[Far65] Farfield, J.M., Gokhale, B.V.: Gold as recombination center in silicon. Solid State Electron. **8**, 685–691 (1965)

[Ful56] Fuller, C.S., Ditzenberger, J.A.: Diffusion of donor and acceptor elements in silicon. J. Appl. Phys. **27**, 544–553 (1956)

[Gal02] Galindo, V., Gerbeth, G., von Ammon, W., Tomzig, E., Virbulis, J.: Crystal growth melt flow control by means of magnetic fields. Energy Convers. Manage. **43**(3), 309–316 (2002)

[Ger79] Gerlach, W.: Thyristoren. Springer, Berlin (1979)

[Ger15] Germain, M., Derluyn, J., Leys, M., Degroote, S.: The material challenge: heteroepitaxial growth of GaN-on-Si, tutorial slides ESSCIC/ESSDERC Conference (2015)

[Gor74] Gorelkinskii, Y.V., Sigie, V.O., Takibaev, Z.S.: EPR of conduction electrons produced in silicon by hydrogen ion implantation. Phys. Status Solidi (a). **22**, K55–57 (1974)

[Gul77] Guldberg, J.: Electron trap annealing in neutron transmutation doped silicon. Appl. Phys. Lett. **31**(9), 578 (1977)

[Haa76] Haas, E.W., Schnoller, M.S.: Phosphorus doping of silicon by means of neutron irradiation with protons. IEEE Trans. Electron Devices. **23**(8), 803–805 (1976)

[Hal96] Hallén, A., Keskitalo, N., Masszi, F., Nágl, V.: Lifetime in proton irradiated silicon. J. Appl. Phys. **79**, 3906 (1996)

[Haz07] Hazdra, P., Komarnitskyy, V.: Local lifetime control in silicon power diode by ion irradiation with protons: introduction and stability of shallow donors. IET J. Circuits Devices Syst. **1**(5), 321–326 (2007)

[Hil15] Hilt, O., Bahat-Treidela, E., Knauer, A., Brunner, F., Zhytnytska, R., Würfl, J.: High-voltage normally OFF GaN power transistors on SiC and Si substrates. MRS Bull. **40**(5), 418–424 (2015)

[Hun73] Huntley, F.A., Willoughby, A.F.W.: The effect of dislocation density on the diffusion of gold in thin silicon slices. J. Electrochem. Soc. **120**(3), 414–422 (1973)

[Jan76] Janus, H.M., Malmros, O.: Application of thermal neutron irradiation with protons for large scale production of homogeneous phosphorous doping of floatzone silicon. IEEE Trans. ED **21**, 797–805 (1976)

[Kao67] Kao, Y.C., Wolley, E.D.: High voltage planar pn-junctions. IEEE Trans El. Dev. **55**, 1409 (1967)

[Kar95] El-Kareh, B.: Fundamentals of Semiconductor Processing Technology. Kluwer Academic Publishers, Boston (1995)

[Ken65] Kennedy, D.P., O'Brien, R.R.: Analysis of the impurity atom distribution near the diffusion mask for a planar p-n junction. IBM J. Res. Dev. **9**, 179–186 (1965)

[Klu11] Klug, J.N., Lutz, J., Meijer, J.B.: n-type doping of silicon by proton implantation. In: Proceedings of the 2011 14th European Conference on Power Electronics and Applications EPE (2011)

[Kra02] Krause, O., Pichler, P., Ryssel, H.: Determination of aluminum diffusion parameters in silicon. J. Appl. Phys. **91**(9) (2002)

[Krü98] Krüger, J., Kim, Y., Subramanya, S., Weber, E.R.: Towards the development of defect-free GaN substrates: defect control in hetero-epitaxially grown GaN by new buffer layer design. Final Report 1997–1998 for MICRO Project 97-202, Berkley. http://www2.lbl.gov/tech-transfer/publications/1461pub.pdf

[Lar51] Lark-Horovitz, K.: Nuclear-bombarded semi-conductors. In: Semiconductor Materials, Proceedings of a Conference at University of Reading. Butterworths, London, 1951, pp. 47–69 (1951)

[Lap08] LaPedus, M.: Industry Agrees on First 450-mm Wafer Standard. EETimes 22 Oct 2008

[Las97] Laska, T., Matschitsch, M., Scholz, W.: Ultra thin-wafer technology for a new 600V-NPT-IGBT. In: Proceedings IEEE International Symposium on Power Semiconductor Devices and IC's, ISPSD '97, pp. 361–364 (1997)

[LiB86] Li, Z., Bradt, R.C.: Thermal expansion of the hexagonal (4H) polytype of SiC. J. Appl. Phys. **60**(2) (1986)

[Lis75] Lisiak, K.P., Milnes, A.G.: Energy levels and concentrations for platinum in silicon. Solid-State Electron. **18**, 533–540 (1975)

[Lut97] Lutz, J.: Axial recombination centre technology for freewheeling diodes. In: Proceedings of the 7th EPE, Trondheim, 1.502 (1997)

[Lut98] Lutz, J., Südkamp, W., Gerlach, W.: IMPATT oscillations in fast recovery diodes due to temporarily charged radiation induced deep levels. Solid-State Electron. **42**(6), 931–938 (1998)

[Mil76] Miller, M.D.: Differences between platinum- and gold-doped silicon power devices. IEEE Trans. Electron. Dev. **23**(12) (1976)

[Mon02] Monakhov, E.V., Avset, B.S., Hallen, A., Svensson, B.G.: Formation of a double acceptor center during divacancy annealing in low-doped high-purity oxygenated Si. Phys. Rev. B **65**, 233207 (2002)

[Mue93] von Münch, W.: Einführung in die Halbleitertechnologie. B.G. Teubner, Stuttgart, Germany (1993)

[Niw08] Niwa, F., Misumi, T., Yamazaki, S., Sugiyama, T., Kanata, T., Nishiwaki, K.: A study of correlation between CiOi defects and dynamic avalanche phenomenon of PiN diode using he ion irradiation. In: Proceedings of the PESC, Rhodos (2008)

[Nov89] Novak, W.D., Schlangenotto, H., Füllmann, M.: Improved Switching Behaviour of Fast Power Diodes. PCIM Europe (1989)

[Ohm72] Ohmura, Y., Zohta, Y., Kanazawa, M.: Electrical properties of n-type Si layers doped with proton bombardment induced shallow donors. Solid State Commun. **11**(1), 263–266 (1972)

[Pic04] Pichler, P.: Intrinsic Point Defects, Impurities, and Their Diffusion in Silicon. Springer Wien, New York (2004)

[Qua08] Quay, R.: Gallium Nitride Electronics. Springer, Berlin Heidelberg (2008)

[Rod05] Roder, C., Einfeldt, S., Figge, S., Hommel, D.: Temperature dependence of thermal expansion of GaN. Phys. Rev. B **72**, 085218 (2005)

[Rys86] Ryssel, H., Ruge, I.: Ion Implantation. Wiley, New York (1986)

[Scl74] Schnöller, M.S.: Breakdown behaviour of rectifiers and thyristors made from striation-free silicon. IEEE Trans. Electron Devices ED **21**, 313–314 (1974)

[Sco82] Schlangenotto, H., Silber, D., Zeyfang, R.: Halbleiter-Leistungsbauelemente - Untersuchungen zur Physik und Technologie. Wiss. Ber. AEG-Telefunken **55**(1–2) (1982)

[Scu89] Schulze, H.J., Kuhnert, R.: Realization of high voltage planar junction termination for power devices. Solid State Electron. **32**(S), 175 (1989)

[Scu02] Schulze, H.J., Niedernostheide, F.J., Schmitt, M., Kellner-Werdehausen, U., Wachutka, G.: Influence of irradiation-induced defects on the electrical performance of power devices. In: ECS Proceedings, 2002-20, pp. 320–335 (2002)

[Scu16] Schulze, H.J., Öfner, H., Niedernostheide, F.J., Laven, J.G., Felsl, H.P., Voss, S., Schwagmann, A., Jelinek, M., Ganagona, N., Susiti, A., Wübben, T., Schustereder, W., Breymesser, A., Stadtmüller, M., Schulz, A., Kurz, T., Lükermann, F.: Use of 300 mm magnetic Czochralski wafers for the fabrication of IGBTs. In: Proceedings of the 28st ISPSD, Prague, pp. 355–359 (2016)

[Sie01] Siemieniec, R., Netzel, M., Südkamp, W., Lutz, J.: Temperature dependent properties of different lifetime killing technologies on example of fast diodes. IETA2001, Cairo (2001)

[Sie02] Siemieniec, R., Südkamp, W., Lutz, J.: Determination of parameters of radiation induced traps in silicon. Solid-State Electron. **46**, 891–901 (2002)

[Sie06] Siemieniec, R., Niedernostheide, F.J., Schulze, H.J., Südkamp, W., Kellner-Werdehausen, U., Lutz, J.: Irradiation-induced deep levels in silicon for power device tailoring. J. Electrochem. Soc. **153**(2), G108–G118 (2006)

[SIL06] Siltronic, A.G.: Float Zone Silicon at Siltronic. www.siltronic.com/int/media/publication/…/Leaflet_Floatzone_en.pdf (2006)

[Ste85] Stengl, R., Gösele, U.: Variation of lateral doping – a new concept to avoid high voltage breakdown of planar junctions. In: IEEE IEDM 85, pp. 154 ff (1985)

[Sue94] Südkamp, W.: DLTS-Untersuchung an tiefen Störstellen zur Einstellung der Trägerlebensdauer in Si-Leistungsbauelementen, Dissertation, Technical University of Berlin (1994)

[Swe83] Swenson, C.A.: Recommended values for the thermal expansivity of silicon from 0 to 1000 K. J. Phys. Chem. Ref. Data. **12**, 179–182 (1983)

[Sze88] Sze, S.M.: VLSI Technology. McGrawHill, New York (1988)

[Sze02] Sze, S.M.: Semiconductor Devices, Physics and Technology, 2nd edn. Wiley, New York (2002)

[Tan59] Tannenbaum, M.: Uniform n-type silicon, U.S. patent 3076732, filled 12 Dec 1959

[Tan61] Tannenbaum, M., Mills, A.D.: Preparation of uniform resistivity n-type silicon by nuclear transmutation. J. Electrochem. Soc. **108**, 171–176 (1961)

[Tan10] Tang, H., Fang, Z.Q., Rolfe, S., Bardwell, J.A., Raymond, S.: Growth kinetics and electronic properties of unintentionally doped semi-insulating GaN on SiC and high-resistivity GaN on sapphire grown by ammonia molecular-beam epitaxy. J. Appl. Phys. **107**, 103701 (2010)

[Tru60] Trumbore, F.A.: Solid solubilities of impurity elements in germanium and silicon. Bell Syst. Tech. J. **39**, 205–233 (1960)

[Tsa83] Tsai, J.C.C.: Diffusion. In: Sze, S.M (eds.) VLSI Technology, McGraw-Hill Book Company, pp. 169–218 (1983)

[Ued05] Ueda, D., et al.: AlGaN/GaN Devices for Future Power Switching Systems. IEEE International Electron Devices Meeting, IEDM Technical Digest, pp. 377–380 (2005)

[Ued17] Ueda, D.: Properties and Advantages of Gallium Nitride. In Meneghini, M., Meneghesso, G., Zanoni, E. (eds.) Power GaN Devices - Materials, Applications and Reliability, Springer International Publishing, Switzerland (2017)

[Ura99] Ural, A., Griffin, P.B., Plummer, J.D.: Fractional contributions of microscopic diffusion mechanisms for common dopants and self-diffusion in silicon. J. Appl. Phys. **85**, 6440 ff (1999)

[Van04] Van de Walle, C.G., Neugebauer, J.: First-principles calculations for defects and impurities: Applications to III-nitrides. J. Appl. Phys. **95**, 3851 (2004)

[Vob02] Vobecký, J., Hazdra, P.: High-power P-i-N diode with the local lifetimecontrol based on the proximity gettering of platinum. IEEE Electron Device Lett. **23**(7), 392–394 (2002)

[Vob07] Vobecký, J., Hazdra, P.: Radiation-enhanced diffusion of palladium for a local lifetime control in power devices. IEEE Trans. Electron Devices **54**(6), 1521–1526 (2007)

[Vob09] Vobecký, J., Záhlava, V., Hemmann, K., Arnold, M., Rahimo, M.: The radiation enhanced diffusion (RED) diode - realization of a large area $p^+p^-n^-n^+$ structure with high SOA. In: Proceedings of the 21st ISPSD, Barcelona, pp. 144–147 (2009)

[Won85] Wondrak, W.: Erzeugung von Strahlenschäden in Silizium durch hochenergetische Elektronen und Protonen, Dissertation, Frankfurt (1985)

[Won87] Wondrak, W., Boos, A.: Helium implantation for lifetime control in silicon power devices. In: Proceedings of ESSDERC 87, Bologna, pp. 649–652 (1987)

[Yim74] Yim, W.M., Paff, R.J.: Thermal expansion of AlN, sapphire, and silicon. J. Appl. Phys. **45**(3) (1974)

[Zie06] Ziegler, J.F., Biersack, J.P.: The stopping and range of ions in matter. [Online]. http://www.srim.org/SRIM/SRIMINTRO.htm. Accessed 1 Mar 2006